

Manual

Data Collection/Information Functions for ERP Batches, MES Batches, Serial Numbers AIP-TRT 8.2

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1 Overview of Data Collection/Information Functions for ERP Batches, MES Batches, Serial Numbers

Purpose

The AIP features contained in this function package make it possible to enter batch-related data directly in production using shop floor terminals or data acquisition PCs.

Integration

Data entered via the AIP terminal can be displayed and/or evaluated in different MOC applications. The collected data can be uploaded via relevant interfaces.

Functions

Order-related data acquisition and posting functions

- Input batches can be entered at the same time as operations are logged on (configurable per operation and workplace)
- Produced output batches can be entered at the same time as operations are logged on
- Changing of input and output batches can be recorded and checked for validity while the operation is running and/or the order is being processed
- Entry of goods receipt batches
- Automatic generation of material movements (incoming goods/outgoing goods) depending on the receipt or consumption of input and/or output batches
- Batch-related collection of quantities and time for produced batches of material
- Entry of ERP or MES batch numbers using the keyboard and/ or barcode
- A validation check to ascertain if documentation is required (documentation obligation) for consumable material when the operation and/or the input batch is logged on
- A ticket/ label is automatically printed in the HYDRA standard format using the assigned printer when a new ERP or MES batch is generated

2 AIP2 Operation

2.1 Special Control and Display Elements on the AIP2

Tables

Uniform selection lists are used in AIP 8.2 posting dialogs:

Filter	
Status	Status
1	PRODUCTION
2	SETUP
3	ORDER SHORTAGE
4	OUT OF MATERIAL
5	OUT OF STAFF
6	START UP
22	COLOUR CHANGE
97	ELECTRICAL FAILURE
98	MECHANICAL FAILURE
99	GENERAL DISTURBANCE

Page 1 • Page 2

If information is available for more than one page, the page numbers are displayed below the table. The current page is highlighted in bold letters. If the user clicks/touches a page, the display directly changes to this page.

If more pages are available than the page numbers displayed, the following buttons can be displayed on the left or right hand side depending on the context (available as of SP10/2016):

-  : If you click this button, the system jumps to the first page of the next page navigation. This means: If *Page 1 ... Page 9* were displayed for the page navigation, the system jumps to *Page 10*.
-  : If you click this button, the system jumps to the first page of the next page navigation. This means: If *Page 10 ... Page 18* were displayed for the page navigation, the system jumps to *Page 9*.
-  : If you click this button, the system directly jumps to *Page 1*.
-  : If you click this button, the system directly jumps to the last page.

You can select an operation using the mouse, touch screen, keyboard (arrow keys: '↑' or '↓'), scanner or by entering it manually.

The content of tables or lists depends on the respective context. Example: When you log on an operation, those operations are available that are included in the sequencing list or planned for the respective workplace or group. When you interrupt an operation, only running operations are available for selection.



Scrolling page by page (up or down) in the table.



Scrolling to the left or right. Only those buttons are activated that make sense for the current situation (context sensitive). This figure shows that scrolling to the left has been deactivated.



Optionally you can display a “table filter” (customization). This is an automatic filter that, once it has been entered, directly affects the table without having to update it. This process is realized through full-text search for (defined) columns. The search is *case-insensitive*.

Virtual keyboard

Using the virtual keyboard, you can enter data manually via touch screen or a connected mouse. The virtual keyboard is displayed automatically as soon as the focus is on an input field. The keyboard layout, which is installed and activated in the Windows language settings, specifies the layout of the virtual keyboard.



Moving the virtual keyboard



Hiding the keyboard for 10 seconds



Switching between the alphanumeric and numeric keyboard



Selecting the keyboard layout (language)



Changing the scaling/size of the keyboard



To move the keyboard, you must configure the driver accordingly (configuration in the control panel of the terminal/PC)!

If you do not want to display the virtual keyboard in general, you must enter the parameter `-t` in the entry `parameters=` of the configuration file `ctaip.ini`.

Date display

AIP supports a country-specific date format in dynamic dialogs. The option "short date" has to be selected in the "regional settings" of the Windows "control panel" of the terminal/PC. Please note:

- Years are always four characters long.
- Months and days are always 2 characters long.
- Allowed separators are: '-' (minus), '\' (backslash) and '.' (dot).
- Blanks must not be included in the "short date" format, i.e. the <BLANK> separator is not allowed.
- The date separator "." (dot) is only allowed in connection with the DD.MM.YYYY format.
- The date format, which might possibly be configured in dynamic dialogs, is ignored.

Examples

- | | |
|-----------------------|------------|
| • English(USA) | MM/DD/YYYY |
| • Danish | MM-DD-YYYY |
| • Customer-specific 1 | YYYY-MM-DD |
| • Customer-specific 2 | YYYY/MM/DD |

- Customer-specific 3 MM/YYYY/DD

Note

If the date format used is other than the permitted formats, a note appears when the program is started and the date format is set to MM/DD/YYYY.

In the status bar, the year format is shortened and displayed only with two characters.

2.2 General description of the posting process on the AIP2

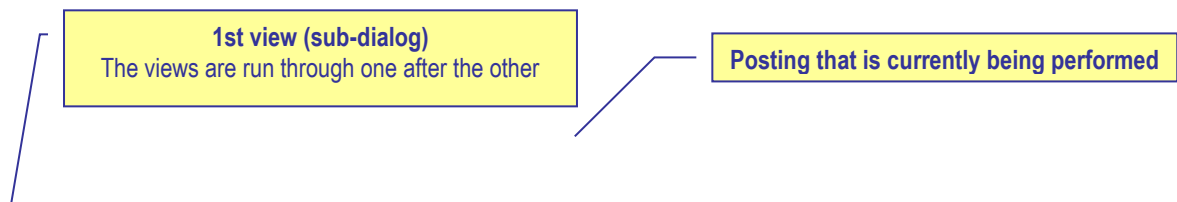
Many AIP posting dialogs are divided into several views (sub-dialogs). These views (sub-dialogs) cover the entire screen so that only one dialog is visible at a time. In a “workflow concept” the user is navigated through the posting dialog step by step. In the following, this process is described using the example *Interrupt operation*. Other posting dialogs are operated in the same way.

The action *Interrupt operation* is performed. To start this action, you click the button *Interrupt* when you have selected an operation:



Interrupt

The dialog *Interrupt operation* opens and the first view (sub-dialog) is displayed. The header displays the function that is currently being executed (here: *Interrupt operation*).



In the first dialog *Enter quantities*, the user can enter the produced yield or scrap quantities. Subject to the active input field, the virtual keyboard is shown or hidden automatically.

Quantities can be entered using the virtual or real keyboard. The user can go to the next field using the tabulator key (which can also be found on the virtual keyboard). When the user has entered all values in the first view, the next view (sub-dialog) can be opened by clicking **Next**.

The **Cancel** button is displayed in all sub-dialogs. Click this button to cancel/close the entire process at any time.

To open the next view (**Select status** in the example), click the **Next** button or another tab (in our example: **Select status** or **Confirmation**). Please note in this context, that no view can be skipped when they are navigated upwards (view 1 → view 2 → view 3). This means: When you are in the first view (*enter quantities*) and you click the third view (*confirmation*), the second view (*select status*) will be displayed first.

Vice versa, when navigating downwards (e.g. from the *confirm* view to the *enter quantities* view), each view can directly be opened by clicking the required tab. In this case, views can actually be skipped. Using the **Back** button, views are opened one after the other (upwards).

As long as the dialog has not been confirmed, the data entered can be changed at any time by scrolling back and forth.

Interrupt operation S61003050100 InjectionMolding at workplace 60610

1 Enter quantities 2 **Choose status** 3 Confirmation

Workplace 60610 ENG21

Actual status OUT OF MATERIAL since 02.10.2015 15:40:10

Filter

Status	Status
1	PRODUCTION
2	SETUP
3	ORDER SHORTAGE
4	OUT OF MATERIAL
5	OUT OF STAFF
6	START UP
15	MAINTENANCE
20	REPAIR
22	COLOUR CHANGE
24	TECHNICAL FAILURE
97	ELECTRICAL FAILURE
98	MECHANICAL FAILURE

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New status 4 OUT OF MATERIAL Estimated duration 0 Minutes

Comment

Cancel Back Go on

mpdv 04.12.15 15:15:00

In the second view *Select status*, you select the workplace status that is set, when the operation has been interrupted. You can select the status from the status list displayed. This list can be restricted using the *Filter* field. Once the required values have been entered, the next view/sub-dialog can be opened by clicking **Next** (in our example it is the last view).

Interrupt operation S61003050100 InjectionMolding at workplace 60610

1 Enter quantities 2 Choose status 3 Confirmation

Operation 0100 InjectionMolding

Prod.Order	S6100305	Target qty.	90000 PCS
Article	SAR-FM0011-K-5013	Yield	1442 PCS
	Mirror housing right, blu	Scrap	33 PCS
		Completion	1%

Workplace 60610 ENG21

New status OUT OF MATERIAL

Entered quantities

Yield	5
Scrap	0

Staff Number

Buttons: Cancel, Back, Interrupt OP

Footer: mpcdv, UK flag, 04.12.15 15:15:45

The sub-dialog *Confirmation* shows a summary of all values entered in the dialog. If the user agrees with the entered data, the **Interrupt operation** dialog can be confirmed, once the badge number has been entered. Then the dialog is sent to the server and posted.

If an input field of the dialog is not completed properly (e.g. a mandatory field is empty), the field is highlighted in red in the respective view and gets the focus. The user can then directly correct the value.



If a workflow dialog is opened, you can click the ESC key to directly exit the dialog. This exit is also possible, if the virtual keyboard is displayed. As a consequence, you cannot use the ESC key to close the virtual keyboard.

3 Main View with Tiles

With the AIP2, the user can switch between the tile design optimized for touchscreens and the list format. By default, the tile layout is shown, which is described in the sections that follow.



To ensure proper processing and posting, terminals with "MDE" operation mode must not be switched off during times without shift.

3.1 Main view – header and footer

Header



The AIP logo is displayed on the top left of the screen, which may be replaced with a customer logo after configuration.

Possible messages are displayed to the right of it (e.g. if a dialog is opened for more than five minutes).

A separate window opens to display error messages that occur during data collection (e.g. validity checks).

Main views

You can assign a maximum of 16 workplaces or machines to the AIP2 terminal. The different workplaces are listed in the order that they were assigned to the terminal on the client. .

In the main view of the AIP2, you can use the button "< Overview" to switch to the icon view of workplaces. In the terminal configuration of the client you can specify whether you want to use the icon view. The sections that follow describe the main view and the icon view.

Footer



CTAIP V8.2.0.0 Server:win2008-7:7 TNR:888

09.09.15 14:47:56

The MPDV logo can be found at the bottom left of the AIP2 terminal. Double clicking the logo opens the info dialog where you can start further administration functions. This dialog closes automatically after approx. 5 seconds.

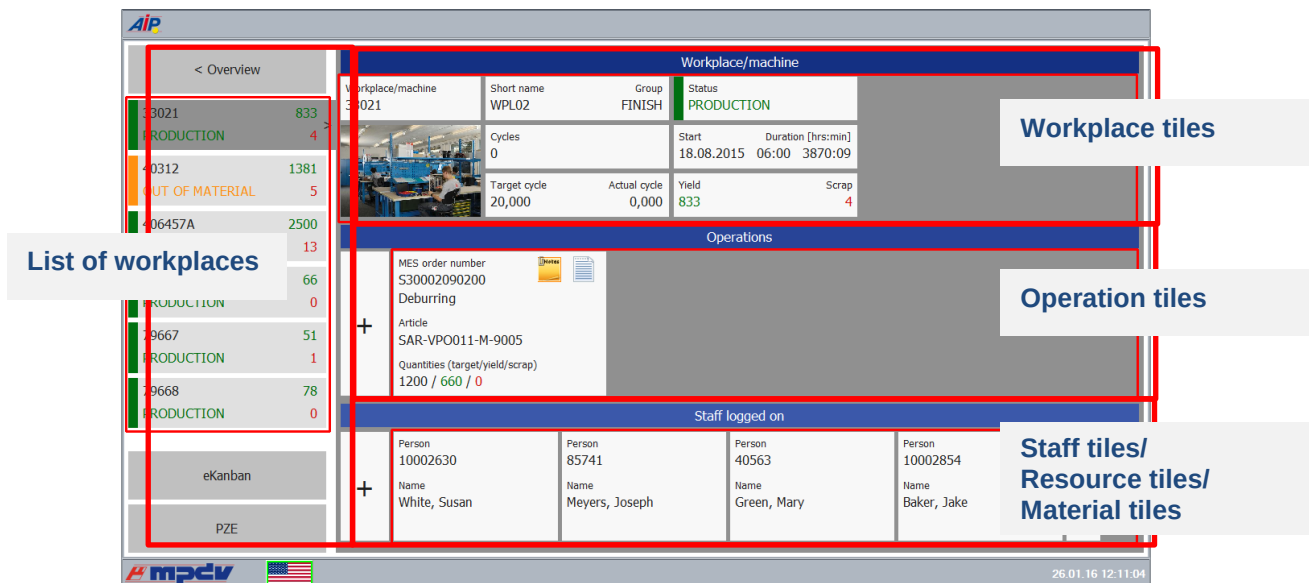
In the middle, further information is displayed: the current terminal status, AIP2 version number, date of the build, IP address of the server and the terminal number.

The current date and time are displayed to the right.

AIP2 statuses

	Network connection has been established. The terminal is ONLINE. Server communication is enabled. All saved data records have been transferred.
	The terminal is sending data to the server.
	No network connection or no connection to the server. The terminal is OFFLINE. Server communication is interrupted. Online functions, such as the display of information, are disabled. But you can still record certain postings. These postings are transferred to the server, once data connection has been re-established.
	Data is being received. The terminal reads files from the server or writes data to the server.
	The terminal is sending saved data records to the server.
	DEMO mode The terminal is in DEMO mode, i.e. server communication is disabled.

3.2 Main view with "tiles"



Please note: The actual display can be different to the above illustration.

Subject to the configurations made, the main view with tiles consists of two or three rows of tiles. While the first two rows of tiles (workplace and operation tiles) are always displayed, it is up to the user whether or not the third row of tiles is shown (optional display). In the configuration of workplaces you can configure for each workplace separately if you want to show the third row of tiles.

Additionally, there is the list of workplaces to the left. Here, you can select the workplaces for which details are displayed on the right-hand side.

List of workplaces

The workplace list shows all workplaces/machines assigned to the terminal. If many workplaces are assigned, swipe to get to the workplaces displayed further down.

This information is shown for the workplaces:

Machine/workplace number

Shows the machine and/or workplace number.

Status

The status is displayed for each machine in color on the left-hand side and also the status text is colored. Coloring is as follows:

- green: production
- yellow: assigned status
- red: not assigned

If the production lock is enabled, an exclamation mark is displayed in the same color as the status.

Quantities

On the right-hand side, the first figure shows the produced yield in green and the red figure shows the produced scrap.



If you have enabled the *Compensate manual quantities* option (e.g. set off scrap against yield) and the machine list also shows shift-related quantities, they will not be updated immediately. The application only updates the quantities, once the lists have been reloaded.

Unit for yield and scrap

If no operation is logged on, the primary quantity unit from the workplace/machine configuration is displayed as unit for yield and scrap. If an operation is logged on, the primary quantity unit of the operation is displayed.

Workplace tiles

The workplace tiles provide the following details:

Workplace/machine

No workplace or machine number is displayed.

Short name / group

The short name of the workplace or machine and the machine group are displayed.

Status

The status is displayed in color and as status text on the left-hand side. Coloring is as follows:

- green: production
- yellow: assigned status
- red: not assigned

If the production lock is enabled, an exclamation mark is displayed in the same color as the status.

Machine image

Shows the picture of the machine stored in the configuration of workplaces.

Clocks

Shows the recorded machine cycles of the current shift.

Start / Duration [hrs:min]

Point in time since the status has been available and the resulting duration at the current point in time.

With BDE workplaces¹, this point in time refers to the last manual status change. With MDE workplaces, it refers to the time when the last status change was identified (for machine connections). It also refers to the point in time when the status was last changed manually or to the time of the last shift change.

Target/actual cycle

Current target and actual cycle of the workplace.

The largest target cycle of all operations logged on to the workplace is shown. The largest target cycle is transferred to the MDE for monitoring.

If the target cycle is smaller than the minimum cycle time, the target cycle is still shown.

If an operation is logged off or interrupted, the largest target cycle of the remaining operations is identified and displayed. After logoff or interruption of the last operation at the workplace, the last target cycle set is still displayed.

If no operation is logged on, the target cycle specified in the machine list is displayed. Thus, even after a restart, the terminal can get the target cycle that last applied.

Yield / Scrap

Yield and scrap quantities of the current shift produced at the machine/workplace.

KPIs: OEE, utilization efficiency, scrap ratio



This function is only available if you enable the extension [aipkpi](#).

Shows the KPIs OEE, utilization efficiency and scrap ratio. The application calculates the KPIs at cyclic intervals (scheduler job "MDE keyfigure calculation"). The KPIs always refer to the current shift. The application calculates the KPIs based on the formulas that are also used for the OEE report and/or the efficiency report on the MOC. The application shows the KPIs with two decimal places. To the right of the KPI, the AIP2 GUI highlights in color if limit values are exceeded or not reached. If you have not defined limit values, the application shows the KPI in gray, otherwise in the color you defined for exceeding/not reaching limit values. For further information on the configuration, refer to the document [MDE KPI Configuration.pdf](#).

¹ An MDE workplace is a workplace that is assigned to a terminal, which runs in the "MDE" operation mode. Otherwise, it is a BDE workplace.



The AIP calculates and updates data at cyclic intervals. This may result in deviations between the collected values and the displayed KPIs.

Linked functions

If you click the workplace status, the dialog for changing the status opens. If you click one of the other tiles, the dialog opens where you can start the functions available for the selected workplace.



The buttons displayed depend on the selected workplace. The button *Lock production status*, for example, is only available for MDE machines.

List of operations logged on

The middle area on the right-hand side shows the logged on operations as tiles. The following data is shown:

MES order number

Order number and operation number of the operation logged on. The combination of these two numbers is the *MES order number*.

Article

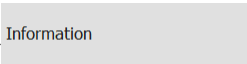
Article defined for the operation.

Quantities (target / yield / scrap)

Shows the target quantity defined for the operation, the produced yield and the scrap. The yield and scrap quantities integrate the counter readings of the available machine connections.

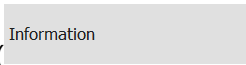


This icon is displayed if at least one note has been recorded for this operation in the graphic planning board of the client that must be shown on the terminal. To display the note(s), click the icon or click the

operation and select the *Information* button ().



This icon is displayed if at least one long text is stored for this operation. Click this icon to display the

long texts or click the operation and select the button *Information* ().

If you click an operation, a screen opens that shows the workplace/operation data. Via this screen, you can also select the operation-related functions.

Operation tiles

In addition to the fields already described, the operation tiles also show the following data:

Comments

This tile shows the user fields 53 and 54 (alphanumeric, 20 characters) of the operation. To edit these fields, you must store a respective user field key for the operation, which includes these two fields.

Completion in %

The bar shows the proportion of "yield", which has been produced until now, compared to the "target quantity".

Since logon (target / yield / deviation)

The production quantity to be expected since the OP has been logged on (depending on the cycle time, partitioning and the time when no production lock has been set for the machine). If the terminal program has been restarted after the OP logon, no value can be calculated.

Calculation:

Target Since Logon = Net Running Time[sec] * Partitioning/Target Cycle[sec/stroke]

Net Running Time: Time since logon while the production lock has not been set. This calculation does not integrate the breaks specified in the shift model or the status times posted to RPA 12 (resource performance account).

Deviation (in percent) between the calculated target quantity since logon and the quantity which has actually been produced "since logon".

Calculation: Deviation[%] = 100% * (Yield Since Logon - Target Quantity Since Logon) / Target Quantity Since Logon.

As of CTAIP 8.2.1.32:

Up to now, the target quantity since OP logon was only calculated if the workplace/machine was assigned to a terminal with operation mode "MDE processing". As of CTAIP 8.2.1.32, the target quantity since OP logon is also calculated if the terminal is configured with operation mode "BDE processing".

If the terminal has been restarted after an operation logon, the target quantity cannot be calculated correctly. To improve transparency, an "*" (asterisk) is shown behind the target quantity (since OP logon) in this case. The asterisk indicates that the target quantity now displayed no longer refers to the time of the operation logon, but to the time of the terminal restart.

While the workplace/machine status 999 is displayed, the target quantity since OP logon is "---". If the status 999 is again changed within a "free shift", the target quantity since OP logon is calculated using the point in time of the OP logon, of the shift start or of the terminal start.

To disable the calculation of the target quantity since OP logon on the terminal, you can use the configuration CalcTargetYieldSinceLogon=0 in the hytnrcfg.ini. The quantity is then displayed using "--".



With workplaces/machines that are configured as "Machining centers", the calculation of the target quantity since OP logon is generally disabled because this calculation contradicts the principles of the machining center.

Planned duration

The field *Planned duration* displays the target processing time of the operation in format [h:mm].

Partitioning

Calculate the displayed partitioning as follows:

TLG _M	Partitioning of the workplace/machine (TLG in mnrlst)
DIV _M	Pulse factor of workplace/machine (IMPFAKT in mnrlst)
TLG _{AG}	Partitioning of the operation (TLG in anrlst)

The application shows the calculated partitioning without decimal places, provided it is an integer value.

In case the partitioning or pulse factor of a machine or an order is 0, calculation is based on the value 1.

Displaying the "3rd list"

The third list is optional. You can configure the third list in the configuration of workplaces. The following lists can be displayed:

- List of staff logged on to the currently selected workplace (BDE)
- List of resources logged on to the currently selected workplace (WRM)
- List of materials/input batches (MPL/TRT) logged on to the currently selected workplace
- List of output batches produced in the currently selected operation (MPL/TRT)

In case you have enabled several lists, you can switch between these lists in the header line that is located above the third list. Activated lists can be selected one after the other.

Maintenance status

If you have purchased the license for the maintenance calendar, the maintenance status is displayed using a yellow or a red field showing a wrench. The color displayed depends on the required maintenance activity.

Calling functions

The functions available are assigned to the relevant objects. Example: The functions *Log person off* and *Log all staff off* are displayed if you click on a person logged on.

3.3 Icon view of workplaces

You can enable this view in the configuration of terminals via the client. Then open this view by clicking the button "< Overview" in the main view. This view shows workplaces in a clear structure and with an image. It shows important information on the single workstations:

Workplace/machine 33021	Workplace/machine 40312	Workplace/machine 406457A	Workplace/machine 406457B	Workplace/machine 79667	Workplace/machine 79668
Status PRODUCTION	Status OUT OF MATERIAL	Status PRODUCTION	Status PRODUCTION	Status PRODUCTION	Status PRODUCTION
Yield 833	Yield 1381	Yield 2500	Yield 66	Yield 51	Yield 78
Scrap 4	Scrap 5	Scrap 13	Scrap 0	Scrap 1	Scrap 0
					

Please note: The actual display can be different to the above illustration.

A colored bar to the left of the image indicates the current status of workplaces/machines:

- green: production
- yellow: assigned status
- red: not assigned

Each tile includes the workplace/machine number, the status (text) of the workplace, the yield and scrap quantity and the image of the workplace.

A colored background with a caliper and/or wrench to the right of the image indicates if an inspection or maintenance is due for the workstation.



If you enable the extension [aipkpi](#), the application also shows the KPIs OEE, utilization efficiency and scrap ratio.

The application calculates the KPIs at cyclic intervals (scheduler job "MDE keyfigure calculation"). The KPIs always refer to the current shift. The application calculates the KPIs based on the formulas that are also used for the OEE report and/or the efficiency report on the MOC. The application shows the KPIs with two decimal places. To the right of the KPI, the AIP2 GUI highlights in color if limit values are exceeded or not reached.



The AIP calculates and updates data at cyclic intervals. This may result in deviations between the collected values and the displayed KPIs.

If you click on a tile, the previously described main view is displayed and the workplace is automatically selected. From there, you can perform the postings for the selected workplace.

Use the option "< Overview" to exit the main view and to return to the icon view.



As part of the advanced configuration options, you can customize the layout of display lists, the displayed data fields and functions. For technical reasons, however, you cannot change the sort sequence of display lists in the main view of the terminal.

As of AIP 8.2.2.28, you can automatically change from the main view with tiles to the icon view of workplaces after a configured time. The configuration `AUTOMATIC-CHANGE-TO-START-DISPLAY` is described in the document [AIP2_Configuration_hytnrcfg.pdf](#).

4 Basic Screen as List View

The new tile design can be disabled for the AIP2 terminal. This chapter describes the basic screen with disabled tiles.

In general, the AIP2 has been designed for entries to be made via touch screen. The corresponding functions can be started, selected or executed by touching the buttons or using the displayed virtual keyboard. Selection lists are provided in many cases, as an alternative to manual entries. Required entries can easily be selected from these lists.

Barcodes can be imported/entered in the current dialog using barcode readers, handheld scanners, or swipe card readers. Subject to the barcode prefix, certain data (e.g. operation data) can directly be assigned to the corresponding input field, without having to focus this input field explicitly.

It goes without saying that mouse and keyboard may also be used.



To ensure proper processing and posting, terminals with "MDE" operation mode must not be switched off during times without shift.

4.1 Basic screens – header and footer

Header



The AIP logo is displayed top left of the screen, which may be replaced with a customer logo after corresponding configuration.

Possible messages are displayed to the right of it (e.g. if a dialog is opened for more than five minutes).

A separate window opens to display error messages that occur during data collection (e.g. validity checks).

Basic screens

A maximum of 16 workplaces or machines can be assigned to the AIP2 terminal. The single workplaces can be found within the list area in the order assigned to the terminal via the client. .

As regards the basic screen of the AIP2 terminal, the user can choose between a tabular view, field-related view and an icon view. This can be configured via the configuration of terminals in the client. The single basic screens are described in the sections that follow.

Footer



The MPDV logo can be found at the bottom left of the AIP2 terminal. Double clicking the logo opens the info dialog where further administration functions can be started. This dialog closes automatically after approx. 5 seconds.

Further information is displayed in the center : the current terminal status, AIP2 version number, date of the build, IP address of the server as well as the terminal number.

The current date and time are displayed to the right.

Terminal status

	Network connection has been established The terminal is ONLINE. Server communication is enabled. All saved data records have been transferred.
	The terminal is sending data to the server.
	No network connection or no connection to the server. The terminal is OFFLINE. Server communication is interrupted. Online functions, such as the display of information, are disabled. But certain postings can be recorded anyway. These postings are transferred to the server, once data connection has been established.
	Data are being received. The terminal reads files from the server or writes data to the server.
	The terminal is sending stored data records to the server.
	DEMO mode The terminal is in the DEMO mode, i.e. server communication is disabled.

4.2 Basic screen “tabular view“

1st list
Workplaces assigned to the terminal

2nd list
List of registered operations

3rd list (optional)
e.g. list of registered staff

The screenshot displays the AIP (Advanced Information Platform) interface. It features three main data tables and several control buttons.

Machines/workplaces table:

Machine/	Workplace	Status	Status since	Note
33021	Manual Workplace Deburin...	PRODUCTION	18.08.15 06:00	
40312	Manual Workplace Deburin...	OUT OF MATERIAL	18.08.15 20:55	
406457A	Mounting Place 1	PRODUCTION	18.08.15 18:40	
406457B	Mounting Place 2	PRODUCTION	18.08.15 15:44	

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Buttons: Modify status, Log on operation

Operations on workplace table:

Article	Order	Operation	Target qty.	Yield	Scrap	N	T
SAR-VPO011-M-9005	S3000209	0200 Deburring	1200 PCS	660	0		

Buttons: Partial confirmation, Interrupt operation, Log off operation

Registered persons table:

Person	Name	First name	Advance logon
10002630	White	Susan	
85741	Meyers	Joseph	
40563	Green	Mary	
10002854	Baker	Jake	

Buttons: Staff work time logging, Staff logging

Footer: H mpcdv, 23.11.15 16:15:15

Subject to the configurations made, the tabular basic screen consists of two or three tables. While the first two tables are always displayed, it is up to the user whether or not the third table is shown (optional).

“Machines/workplaces” table

The upper table shows the workplaces assigned to the terminal. The following columns are displayed.

Machine/workplace

The machine or workplace number as well as a description are displayed.

Status

The status is highlighted in color and the status text is shown. Coloring is as follows:

- green: production
- yellow: assigned status
- red: not assigned

Status since

Point in time since the status is available.

For ADE workplaces² the point in time refers to the last manual status change. For MDE workplaces it refers to the time when the last status change was identified (for machine connections). It also refers to the point in time when the status was changed manually most recently or to the time of the last shift change.

Please note:

It is indicated here if the "lock production status" function is enabled for the machine/workplace.

Below the first list there is a row including the function buttons mainly relating to machines/workplaces. These functions are described in more detail in the sections that follow.



By way of "customizing" services it is possible to adapt the layout of the display lists, displayed data fields, sort sequences, etc. according to the customer's requirements. For technical reasons, however, the sort sequence of display lists may not be changed in the basic screen of terminals. The software does not allow it.

Provided that the "compensate manual quantities" option (e.g. set off scrap against yield) is enabled and the machine list also shows shift-related quantities (no default setting), they will not be updated immediately. Quantities are only updated once the lists have been reloaded.

"Operations at workplace" table

The second table shows the operations currently logged on to the selected workplace. The following columns are displayed:

Article

Article defined for the operation

Order and operation

Order number and operation number of the registered operation. Together they build the *MES order number*.

Target quantity

Target quantity defined for the operation.

² We talk of an MDE workplace if this workplace is assigned to a terminal, which runs in the "MDE" operation mode. In any other case, it is an BDE workplace.

Yield

Yield already produced for this operation. The counters of possible machine connections are considered as well.

Scrap

Scrap quantity already produced for this operation. The counters of possible machine connections are taken into account as well.

N

It is indicated here if a note visible on the terminal has been recorded for this operation in the graphic planning board of the client. The note(s) is/are displayed by clicking the OP info button ().

T

If a long text is defined for this operation it is indicated here. The long text is displayed using the OP info dialog (button ).

Below the second list there is a row that mainly includes function buttons relating to operations.

"3rd list" table

The third list is optional and may be configured. Information displayed in this list depends on the workplace configuration.

The following lists can be displayed:

- List of staff logged on to the currently selected workplace (BDE)
- List of resources logged on to the currently selected workplace (WRM)
- Materials/input batches logged on to the currently selected workplace (MPL/TRT)
- List of output batches produced in the currently selected operation (MPL/TRT)

The buttons below the third list (to the left) allow switching between these lists.

Please note

The staff logged on displayed in the third list is identical to the list displayed in the dialog "F5 staff logged on...". Selecting a person in the third list does *not* affect the selection of the operation in the list of OPs running at the workplace. Therefore, it neither affects pre-assignment of the operation in the corresponding posting dialogs.

Toolbar in the basic screen

A toolbar, which may be customized, is assigned to each list included the basic screen. This makes the purpose of a function clear to the user. The “partial upload/confirmation” function can be found below the list of registered operations.

In fact, the toolbar may include several “tabs”, which can be made visible by scrolling to the right/left at the right/left end of the toolbar. A posting dialog (e.g. *change partitioning*) can be opened by clicking the corresponding button.

Please note

The displayed buttons depend on the context defined by the respectively selected workplace. Thus, the displayed buttons may vary when selecting another workplace/machine.

4.3 Basic screen "machine overview"

If the “change view” button is clicked in the basic screen, the view changes to the following presentation:

Toolbar of the assigned machines

Machine information

Order information

Workplace/Machine	33021	Status	PRODUCTION	
Short description	WPL02	Start	06:00	18.08.2015
Group	FINISH	Duration [hh:min]	2338:16	
Part.	1	Yield	833	PCS
Target cycle [sec/cycle]	20	Scrap	4	PCS
Actual cycle [sec/clock]	0	Cycles		
		Note		

Operation	S30002090200	Target qty.	1200	PCS
Article	SAR-VPO011-M-9005	Yield	660	PCS
Article Designation	Mirror housing right,	Scrap	0	PCS
Comment 1				PCS
Comment 2		Yield since logon	0	PCS
Planned duration	06:40	Completion	55%	

This presentation gives detailed information on a single machine, whereas the above toolbar still provides an overview of all assigned machines and workplaces.

The presentation consists of three sections:

Toolbar of the assigned machines

The color indicates the current status of all assigned workplaces. Coloring is as follows:

- green: production
- yellow: assigned status
- red: not assigned

It is possible to switch between machines by pressing a machine icon (requires a touch screen). Using the keyboard, the active machine can be selected by the arrow keys. To do this, the toolbar must be active.

Workplace/machine information

This display area shows information relating to workplaces/machines and shifts about the currently selected workplace.

Order information

This display area shows information on the registered order/OP. If several orders/OPs are logged on to the workplace, then extra arrow buttons are displayed. It is possible to switch between individual orders/OPs using these arrow buttons.

Notes on selected fields of the machine overview

Unit for yield and scrap

Provided that no operation is logged on, the primary quantity unit from the workplace/machine configuration is displayed as unit for yield and scrap. If an operation is logged on the primary quantity unit of the operation is displayed.

Partitioning

The displayed partitioning is calculated as follows:

$$\text{Partitioning} = \frac{TLG_M}{DIV_M} * \left[\frac{TLG_{OP1}}{DIV_{OP1}} + \frac{TLG_{OP2}}{DIV_{OP2}} + \dots \right]$$

TLG _M	Partitioning of the machine (TLG in mnr.lst)
DIV _M	Pulse factor of the machine (IMPFAKT in mnr.lst)
TLG _{OPi}	Partitioning of the individual operation (TLG in anr.lst)

The resulting partitioning is displayed without decimal places, provided it is an integer value. Otherwise, 3 decimal places are shown.

In case partitioning or pulse factor of a machine or an order is 0, calculation is based on the value 1.

Having logged off all OPs, the machine continues working with the partitioning of the machine.

Target cycle

The largest target cycle of all operations running at the machine is always displayed in the machine overview of the terminal. If this OP is logged off the largest target cycle of the remaining OPs will be displayed.

In case no OP is logged on, the target cycle from the machine list is displayed. Thus, even after a restart, the terminal can get the target cycle that applied at last.

The largest target cycle is also transferred to MDE for monitoring.

Comment 1, comment 2

These two fields show the user fields 53 and 54 (alphanumeric with 20 characters) of the operation. To be able to edit these fields, a corresponding user field key containing these two fields must be defined for the operation.

Target since logon

The production quantity to be expected since the OP has been logged on (depending on the cycle time, partitioning and the time while the production status was not locked for the machine). No value can be calculated, in case the terminal program has been restarted since the OP was logged on.

Calculation:

$\text{TargetSinceLogon} = \text{NetRunningTime}[\text{sec}] * \text{Partitioning} / \text{TargetCycle}[\text{sec/stroke}]$

NetRunningTime: Time since logon while the production lock has not been set. This calculation does not take into account any breaks defined in the shift model or status times posted on RPA 12 (resource performance account).

Deviation [%]

Deviation (in percent) between the expected target quantity since logon and the quantity which has actually been produced "since logon".

Calculation: $\text{Deviation}[\%] = 100\% * (\text{YieldSinceLogon} - \text{TargetSinceLogon}) / \text{TargetSinceLogon}$

Completion

The bar represents the proportion of "yield", which has been produced until now, compared to the "target quantity".

Machine icon:

Provided that the WRM-WTK license has been purchased, the machine icon may be replaced with a picture showing a yellow or red oilcan. It all depends on the required maintenance activity:



4.4 “Machines as icons” basic display

This view can be configured as the default display in the configuration of terminals at the client. It has the advantage that the user can tell from a distance whether or not all machines are in the “Production” status.

All MDE machines have their own buttons colored according to the corresponding status:

- green: Production
- yellow: Assigned status
- red: Not assigned

The button includes details on the workplace/machine number, the registered operation, the yield and scrap quantities as well as the status (text) of the workplace/machine.

If a button is touched, the “machine overview” basic display is shown for this workplace. From there, postings for the selected workplace can be performed using the standard buttons.

By clicking the “symbol” button (if configured) the view changes from the “machine overview” to the “icon view of machines”.

5 Barcode Input with Prefix

A barcode is interpreted as prefix barcode if the third character is a dot. In this case the first two characters identify the barcode type. The actual barcode starts with the fourth character.

ID+ Prefix	Example	Comment --> processing
-----	-----	---- General ---
00.	00.ABC123	Data not defined, → 00. will be deleted and "ABC123" will be passed on to standard processing
01.	01.OK 01.ESC	Action barcode → Dialog cancelled or ended with OK button or Esc button.
-----	-----	---- HYDRA-ADE + HYDRA-LLE + HYDRA-MDE ---
10.		(combined) Order/sequence/OP number → acronym <ANR>
11.		Order (header) → acronym <AUNR>
12.		Sequence → acronym <AFOLG>
13.		OP → acronym <AGNR>
14.		Sub-order number -> Acronym <UAGNR>
22.		Split no. → acronym <SPLNR>
15.		Upload/confirmation number → Acronym <RMNR>
16.	16.EXTRUDER-7 16,200	Machine → Acronym <MNR> → Passed on to dialog with MNR=EXTRUDER-7 or MNR=200
17.	17.1 17.1001	Machine status → Acronym <MST> → Passed on to dialog with MST=1 or MST=1001
18.	18.1 18.1001	Scrap reason → Acronym <EGG:AUS> → Passed on to dialog with EGG:AUS =1 or EGG:AUS=1001
19.	19.1 19.1001	Deviation reason → Acronym < EGG:GUT > → Passed on to dialog with EGG:GUT =1 or EGG:GUT=1001
20.	20.1 20.MF	Operator position → Acronym <BPOS> → Passed on to dialog with BPOS =1 or BPOS = MF
21.		Wage and premium indicators → Acronym <LPKZ>
-----	-----	----HYDRA-WRM + HYDRA-DNC + HYDRA-PDV + HYDRA-MPL ---
40.	40,100 40.MONTAGE	Destination → Acronym <ZLO> → Passed on to dialog with ZLO=100 or ZLO= MONTAGE
41.	41.KARTON 41.KISTE	Transport unit → Acronym <TPE> → Passed on to dialog with TPE = KARTON or TPE = KISTE
42.		Batch number → Acronym <CNR>→→
43.		Throughput batch number → Acronym <DLL>→→
44.		Alternative batch number → Acronym <CNR:ALT1>→→
45.		Alternative batch number → Acronym <CNR:ALT2>
46.		Alternative batch number → Acronym <CNR:ALT3>
47.		Alternative batch number → Acronym <CNR:ALT4>
48.		Alternative batch number → Acronym <CNR:ALT5>
49.		Alternative batch number → Acronym <CNR:ALT6>
-----	-----	---- Mainly for the HYDRA-PZE module ---

ID+ Prefix	Example	Comment --> processing
50.		Badge number → Acronym <KNR>
51.		Personnel number → Acronym <PNR>
52.	52.EDV 52.VERTRIEB	Cost center → Acronym <KST> → Passed on to dialog with KST=EDV or KST=VERTRIEB
53.	53.1 53.1001	Absence reason → Acronym <FGR> → Passed on to dialog with FGR=1 or FGR=1001
-----	-----	---- Customer-specific barcodes ---
90.		
...		

In case barcodes are required, which actually have a dot at the third place (e.g. if the machine/workplace number has a dot as third character), it is possible to define an alternative indicator for barcode prefixes in the HyTnrCfg.ini terminal configuration, e.g.

[Terminal->USR 0]

BarcodePrefixChar=\$



If another prefix is actually required, the respective barcode font in use must be able to represent this prefix.

Examples for barcodes

Barcode printing: Font "Codedreineun" and prefix "."

Prefix	Barcode	Raw data
10.	*10.123456780100*	ANR = 123456780100
	10._ABCD12340100	ANR=_ABCD12340100

Prefix	Barcode	Raw data
11.	*11.12345678*	AUNR = 12345678
12.	*12.01*	AFOLG = 01
13.	*13.0100*	AGNR = 0100
14.	*14.0000*	UAGNR = 0000
15.	*15.123456789012345*	RMNR = 123465789012345
22.	*22.02*	SPLNR = 02

Prefix	Barcode	Raw data
16.	*16.123456*	MNR = 123456
17.	*17.1122*	MST = 1122
18.	*18.1234*	EGG:AUS = 1234
19.	*19.123456789*	EGG:GUT = 132456789
20.	*20.13*	BPOS = 13
21.	*21.1221*	LPKZ = 1221

Prefix	Barcode	Raw data
50.	*50.1337*	KNR = 1337

5.1 Configuration of customized barcode prefixes

Section [barcode]	
BarKenn90=SAPCNR BarKenn91=EGR:GUT	The barcode prefixes 90...99 can be assigned here according to the customer's requirements. This means, if a barcode with the relevant prefix is used, it will be transferred to the dialog along with the assigned ID. Then the barcode has the following structure: <Prefix>.<Net barcode> e.g.: "90.12345" → SAPCNR=12345

Firmly assigned barcode prefixes:

10:ANR 53:FGR

11:AUNR

12:AFOLG

13:AGNR

14:UAGNR

15:RMNR

16:MNR

17:MST

18:AUSGRD

19:AGGGRD

20:BPOS

21:LPKZ

22:SPLNR

40:ZLO

41:TPE

42:CNR

43:DLL

44:CNR:ALT1

45:CNR:ALT2

46:CNR:ALT3

47:CNR:ALT4

48:CNR:ALT5

49:CNR:ALT6

50:KNR

51:PNR

52:KST

6 AIP2 -Local Configuration

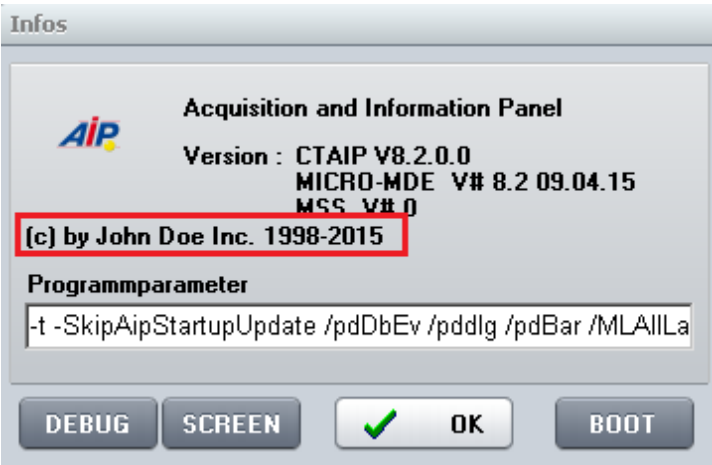
6.1 Local Configuration ctaip.ini

The most important hardware and system settings are defined for each terminal in the CTAIP.INI file of the C:\MPDV\AIP2 directory.



Changes to the configuration file ctaip.ini are only enabled after rebooting the terminal software.

Layout configuration

Entry	Comment
Section [system]	
CompanyInfo=(c) by John Doe Inc. 1998-2015	Overwrites the copyright text in the "About" dialog 
parameters= ... -t +t	The virtual keyboard can be disabled/enabled, irrespective of the terminal type.
DEMO_ALL=ON	The information in the lower bar are normally hidden in demo mode. This option, however, causes all pieces of information to be displayed even in demo mode (version number and date, server, terminal number, online lights).
parameters= ... - SkipDynDlgExtraction...	Internal option preventing the extraction of dialog fields/buttons when updating dialog data. (Only applies if WF directory ".\spool\swf.\$\$\$\" is available)
parameters= ... - SkipAipStartupUpdate...	Internal option preventing the configuration files (*.ini,*.cfg,*.cfi) of the module DLLs (packets*.dll) and application DLLs (functions*.dll) from being synchronized when starting the terminal.
VirtScreenSize=640	All windows are started with the indicated resolution

Entry	Comment
VirtScreenRatio=16:9	The display ratio remains if the configuration VirtScreenSize is used to reduce the window. The width-to-height (aspect) ratio can be changed using VirtScreenRatio. Consequently, the width-to-height ratio 16:9 can be tested with a 4:3 monitor and vice versa. The value can be configured as "16:9" or as floating point value (e.g. 1,77777). Note: Information screens are specifically scaled if monitor screens in portrait format (e.g. 9:16) are used. This special scaling can be disabled in hytnrcfg.ini: [Tnr Konfiguration 0]->PortraitAlignment=off
Section [SKIN]	Configurations for the terminal's skin (see ctaipplay.ini) The online configuration of the terminal can be reached by the info dialog (click MPDV icon). ALT+F1 opens the control elements for the skin. Please note: Only in the design/GUI of AIP 8.1 and/or the dynamic dialogs
Saturation=0	Skin configuration / Default = 0
Hue=0	Skin configuration / Default = 0
Name=mpdv	Skin to be used / Default = mpdv
Active=false	Skins can only disabled in ctaip.ini! Default = true

Entry	Comment
Section [system]	
Usr=21	Distinct terminal number
Hostname=192.9.200.24	Internet address of the server
Offlinetimeout=600	The interval after which online access should be attempted the next time. The interval is specified in seconds
Showcursor=on	Show or hide mouse pointer in terminal application: on: mouse pointer active off: mouse pointer inactive
Loadfile=ctnet\win\aip2.txt	Configuration file to download the application from the server. The paths is relative to the installation directory of the system.
Watchdog=on	ON: Watchdog is activated OFF: Watchdog is not activated

Entry	Comment
Demo=off	'on': Offline demo mode; always off in the production environment!
parameters=-t	The -t parameter switches off the virtual keyboard.
TMOUT_C=xxx	Timeout for CONNECT to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
TMOUT_S=xxx	Timeout for SEND to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
TMOUT_R=xxx	Timeout for RECEIVE of the server If not specified, default = 120 seconds
TMOUT_F=xxx	Timeout for FILESERVER operations to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
Section [barcode]	Configuration of customized barcode prefixes.
BarKenn90=MNR4 ... BarKenn99=ANR3	BarKenn90 > defines the prefix (here: 90); The ID from the dialog (= acronym) is assigned.
Section [comports]	
com1=0 com2=MSS com3=BAR Com3=LEGIC Com4=RFLESER	Assignment of serial interfaces to connected devices: MSS – machine interface BAR, LEGIC, RFLESER – various reading devices
Section [MSS-INIT]	
MSS_DIALOG=10	If the terminal is switched off longer than 15 minutes, a dialog is displayed on terminal restart. The user must then decide whether the counter pulses, which were recorded when the terminal was closed, are posted or discarded. The dialog closes automatically with "Yes" after an entered time has elapsed; in this case the counting impulses are posted. This value configures the time in seconds the dialog is open. If the terminal is switched off for less than 15 minutes, no dialog is opened; the counting pulses recorded in the switch-off phase are accepted and posted without confirmation. Please note: The value can also be configured in hytnrcfg.ini. Entries in the hytnrcfg.ini file take priority.

Entry	Comment
MSS_FILEAGE_MIN=5 MSS_FILEAGE_OVERTIME=delete	Additional configuration for MSS_DIALOG If the backup file for counting pulses of the terminal is older than 5 minutes, no dialog is opened and the back up file deleted. Quantities recorded at the time when the terminal was closed are not used/posted. Please note: The value can also be configured in hytnrcfg.ini. Entries in the hytnrcfg.ini file take priority.
Section [ext. software]	
Button=Editor WindowName=Editor SearchParts=On	Configuration of the button in the top line: A previously started program can be brought to the foreground at the push of a button. Button: button caption WindowName: Name of the program (e.g. from the taskbar). SearchParts=On: Parts of WindowName are sufficient SearchParts=Off: WindowName must be entered completely. The option "SearchParts=On" is recommended for programs such as MSWord that change the title bar subject to the document that is currently being loaded.
ProgFileName=c:\Programme\wincmd\Wincmd32.exe	The program that is started if the program mentioned above cannot be called to the foreground.
AutoStart=on	This option starts the program (ProgFileName) when starting the terminal program.
Section [DLL]	
BusDLL=PCC.EXE	PCC.EXE must be entered here if the terminal is configured with the option "operated as HYDRA-MDE terminal". If necessary, the AIP2 enters this value independently upon starting the program.
Section [PDV]	
PDVProtokoll=ON	Enables PDV logging (prot_pdv.txt) (by default=OFF)
PDVIOPODir=c:\IOPSim\	Path for DNC
Section [CAQ]	
;Supported barcodes	
FieldWNRBarcodeOnly=Y	If this entry is set the tool number may only be entered using a scanner.
FieldNestBarcodeOnly=Y	If this entry is set the cavity number may only be entered using a scanner.
FieldNummBarcodeOnly=Y	If this entry is set the number may only be entered using a scanner.
FieldKNRBarcodeOnly=Y	If this entry is set the badge number may only be entered using a scanner.

Entry	Comment
BarcodeWNR=	This field specifies which acronym is entered into the tool number field by the scanner.
BarcodeNest=	This field specifies which acronym is entered into the cavity number field by the scanner.
BarcodeNumm=	This field specifies which acronym is entered into the number field by the scanner.

6.2 PNG – Files / Bitmaps

The use of PNC files is recommended by MPDV. By default PNG files have a size of 24 x 24 px.

6.2.1 File pict.zip

The file "pict.zip" is updated by the installation tool "inst32.exe" while downloading and includes all default PNG files.

The default PNG files can be overwritten in the file pict_cust.zip. Several PNG files have the extension ".small.png" (e.g. aip.small.png). These PNG files are used with a screen resolution of 640x480.

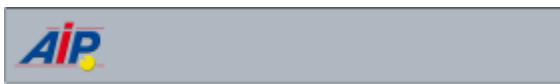
6.2.2 File pict_cust.zip

The file "pict_cust.zip" is loaded from the server directory (e.g. \<serverDir>\1\custom) when starting the program (as is the case for the hycust.mld).

Customized PNG files may be stored in this file and loaded by the AIP2 terminal. Default PNG files may also be "overwritten".

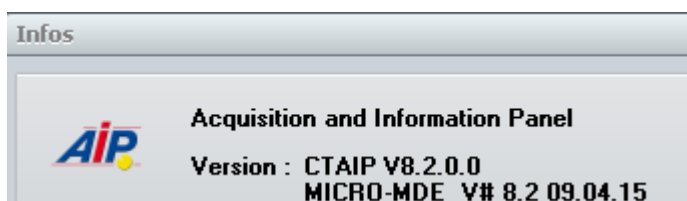
Please note: file sizes are not adjusted.

Customize header



The AIP icon displayed in the header can be replaced by storing a separate AIP.png file in the pict_cust.zip file.

This AIP icon will also be replaced in the "About" dialog.

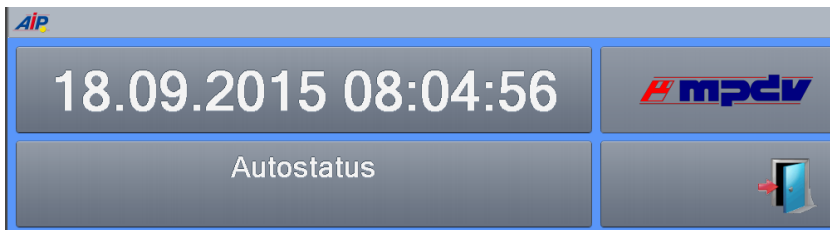


Customize footer



The MPDV icon displayed in the footer can be replaced by storing a separate company.png file in the pict_cust.zip file.

Customize PZE dialog



The MPDV icon displayed in the PZE dialog can be replaced by storing a separate pze_mpdv.png file in the pict_cust.zip file. In case the PZE terminal is operated with a screen resolution of 640x480, a customized pze_mpdv.small.png file has to be integrated in the pict_cust.zip file.

6.3 Multilingualism (*.mld files)

The below-mentioned files are required for the translation of the application. The table shows the priorities for the translation, ownership and the relevant storage locations.

Priority	File	Server directory	Owner	Description
1	ctaipkd.mld	./ctnet/win/aip2/custom	Customer	Customer-specific translations
2	hycust.mld	./1/custom/	MPDV	Customized translations by MPDV. The file is used by the AIP2, CTAIP and CTWIN terminal.
3	ctaip.mld	./ctnet/win/aip2	MPDV	Standard translation file

7 AIP2 - Central Configuration File hytnrcfg.ini

You can use this file as central place to store different configurations for all or for separate terminals.

For each section, a general version is available

[section 0].

The entries included in this section can be overwritten by entries in a terminal-specific section

[section <TNR-USER>]

<TNR-USER> = UserNo = terminal number + 2000 e.g. 2010,2101,..) for exactly one terminal/UserNo



The file hytnrcfg.ini is loaded from the server on every terminal start.

Section / Entry	Comment
	→→→
[Tnr Konfiguration 0]	
FollowExternStatus=on	Transfer of machine statuses when reloading machine list. Useful if status change is set by PDM or another terminal
[Terminal->Installation 0]	
InstallFonts=on	If set to "off", fonts are not installed during restart. ON=DEFAULT
OnlyInstallFontsAfterDownload=false	If "InstallFonts=on": If true, then fonts are only installed directly after a download. If false, then fonts are installed every time the terminal is restarted. (false = DEFAULT)
[Terminal->USR 0]	

Section / Entry	Comment
AttachedApplication=First	Displaying documents of OP info: With this configuration, the system first checks whether or not an application is linked in Windows that matches the file extension of the document. This application is then used to display the document. If no link is available, the viewers configured in ctaip.ini (→ [ext. software]) and internal viewers are used. If an extension is completely unknown, the system tries to display the document as text . Different settings are possible: First → search for linked application first AfterUserViewer → If a UserViewer is configured, this one overrides the linked application (also applies to ExcelViewer, WordViewer and PowerpointViewer) Last → Only if no ctaip.ini assignment is found for the file extension, then the system searches for a linked application (default). Off → The system does not search for a linked application.
HTTPBrowser=standard	Display of documents (via OP info): If documents are configured with a path of schema "http", the file is not downloaded to the terminal, but the link is transferred to a browser. The default browser for the terminal is htmview3.exe, as this one can be operated by touchscreen. If this entry is set, the default browser configured in Windows is used.
SupressErrorMessage=70012	Suppress message "material is not planned"
MSS_DIALOG=10	If the terminal is switched off longer than 15 minutes, a dialog is displayed on terminal restart. The user must then decide whether the counter pulses, which were recorded when the terminal was closed, are posted or discarded. After a configurable period of time, the dialog closes automatically with "Yes" (Yes, posting of pulses). This value configures the time in seconds the dialog is open.
MSS_FILEAGE_MIN=5 MSS_FILEAGE_OVERTIME=delete	If the backup file for counter pulses on the terminal is older than 5 minutes, then no dialog is opened and the backup file deleted. Quantities recorded at the time when the terminal was closed are not used/posted.
[SignatureRecording->User 0]	
ManualBadgeInput=true	This configuration specifies whether or not the field <i>User</i> can be edited on the terminal (by default: no editing) true → activates keyboard input for field <i>User</i> on the terminal

Section / Entry	Comment
Transparency=255	The display of the signature dialog can also be transparent. 255 → Signature dialog is 0 % transparent (not transparent) 1 → Signature dialog is 99% transparent (maximum transparency) (Default = 155)
ShowPosition=TR	You can change the place of the signature dialog: TL Top – Left TM Top – Middle TR Top – Right ML Middle – Left MM Middle – Middle (Default) MR Middle – Right BL Bottom – Left BM Bottom – Middle BR Bottom – Right
USE_SERVICE_ACCOUNT=1	0 (default) SSO: ServiceAccount is not used (requirement: the terminal must be started with the domain "user" (SSO)). Note: ServiceAccount=1 can only be used if all users are in the "root" domain. SubDomain users are not supported.
SIGNATURE_1_USER_TYPE=REPORTING_USER_READONLY	REPORTING_USER_READONLY The user identification using the Windows user is activated. The Windows user is then preassigned in field <i>User</i>. The <i>User</i> field is read-only. Requirement: The "SSO" option must be enabled for all reporting users. Otherwise, successful authentication is not possible. REPORTING_USER_CHANGEABLE The user identification using the Windows user is activated. The Windows user is then preassigned in field <i>User</i>. The <i>User</i> field can be edited. Requirement: The "SSO" option must be enabled for all reporting users. Otherwise, successful authentication is not possible.

Section / Entry	Comment
SIGNATURE_1_LOGON_TYPE=HYDRA	"" / Not set / "EMPTY" There is also an alternative login procedure. HYDRA The user identification using the Windows user is locked. You must enter a user for identification. Requirement: All reporting persons must have been created as users and the option "SSO" must not be enabled for the users. Otherwise, successful authentication is not possible. ACTIVEDIRECTORY The TAB for the user identification via <i>User</i> is locked. The Windows user must be used for identification purposes. Requirement: The "SSO" option must be enabled for all reporting users. Otherwise, successful authentication is not possible. MIXED_BUT_UNIQUE The setting of option "SSO" specifies whether the user login or the Windows login is used. "SSO" enabled → Windows only "SSO" disabled → user only
SIGNATURE_2_LOGON_TYPE=HYDRA	Identical to SIGNATURE_1_LOGON_TYPE (see above)
ExtendedSignatureRecording=true	Used for signatures on the terminal with quality data collection.
[MDE/Blade Configuration 0]	
CONVERT-TO-ANSI- FILE=<list1 list2>	Configuration of the files that are provided from the AIP to the MDEB2 blade in ANSI format when a combined operation is available. The following lists are transferred by default if the entry is not available. counters.lst schicht.lst mnr.lst mstat.lst anr.lst pnr.lst If you want to transfer further lists, you must specify the standard lists and the additional lists.

7.1 Layout configuration

Entry	Comment
Section and/or [terminal configuration 0] [terminal configuration 2XXX];	(general configuration) (2XXX terminal-specific configuration)
AUTO-CONFIRM-UHR-ERROR-MESSAGE=TRUE	This setting specifies that in case of an error that occurred reading the clock (e.g. when activated after standby

Entry	Comment
	mode), the time is transferred without confirmation dialog and the terminal time is later synchronized with the server time via PDM command.
SUPPRESS-MAXIMUM-NUMBER-OF-MACHINES-WARNING=ON	Suppresses the warning after restart of terminal if more than 32 machines are assigned to the terminal (static/dynamic). (Default=OFF)
CalcTargetYieldSinceLogon=2	CalcTargetYieldSinceLogon=1 The duration is calculated from the total runtime since login (all statuses) minus the configured shift breaks. CalcTargetYieldSinceLogon=2 The duration is calculated from the total runtime since login (all statuses). Defined breaks are not used and are not deducted. (default value)
Section [QRD-PRINTER->TICKET 0] ;(general configuration) [QRD-PRINTER->TICKET 2xxx] ;(2XXX configuration for a specific terminal)	
COMPLETE-ABSENCE-OF-LOCAL-MNR-DATA-FOR-EVENT=< Events >	Reloads the machine row for the configured <Events> , if it is not available locally ➔ This configuration might be required for a group workplace without machine assignment.
COMPLETE-ABSENCE-OF-LOCAL-ANR-DATA-FOR-EVENT=< Events >	Reloads the order row for the configured <Events> , if it is not available locally ➔ This option has been implemented to access order data in the master data, e.g. when logging on orders.
COMPLETE-...-EVENT=< Events > COMPLETE-...-EVENT=#ALL# COMPLETE-...-EVENT=A_AN A_P_AN	Explanation on the configuration of <Events> ➔ Using <#ALL#> the row (ANR/MNR) that is not available is reloaded for any event. ➔ <A_AN A_P_AN> restricts reloading of information to specified events. The ID <DLGFAM> is preferred to the ID <DLG> in order to identify the <Event>.
Section [AIP2 Initialization 0]	
XML-GUI=OFF	Disables the new AIP2 design and uses the AIP 8.1 design.
CTWIN-STYLE=ON	Activates the GUI that is similar to CTWIN on the AIP2. The two button bars are shown below the two lists just like on the AIP 8.1.
CTWIN-BUTTON-LAYOUT=ON	If the option CTWIN-STYLE=ON is additionally set, the two button bars are displayed at the bottom of the screen.
AUTOMATIC-CHANGE-TO-START-DISPLAY=30	As of AIP 8.2.2.28: If this option is set, the display automatically changes to the main view after the configured time if no other interaction was performed in the meantime. The changing display is configured via the option <i>Show machine/OP</i> in the <i>Terminal configuration</i>, tab <i>MF functions</i>. - List: - Change from the detail views or function menus (operation, person, resource, etc.)

Entry	Comment
	○ Change to the main view with "tiles" • Icons: ○ Change from the detail views or function menus or from the main view with tiles ○ Change to the icon view of workplaces The configuration is specified in seconds. Do not specify less than 10 seconds. An automatic change to the start screen is not made if an input dialog is open.

8 Order Postings for Operations subject to Batch Management

Overview

In addition to logging the actual operation on, operations subject to batch management also enable to log on relevant input batches.

Configuration

These system settings must be carried out to generally use operations that are subject to batch management.

Basic screen

Basic terminal screen when a machine is assigned in batch mode:

The screenshot displays the AIP terminal interface. On the left, a list of machines is shown with their status and counts. The main area is divided into sections for 'Workplace/machine', 'Operations', and 'Staff logged on'.

Workplace/machine			
Workplace/machine 33021	Short name WPL02	Group FINISH	Status PRODUCTION
Cycles 0		Start 18.08.2015	Duration [hrs:min] 06:00 2338:22
Target cycle 20,000	Actual cycle 0,000	Yield 833	Scrap 4

Operations	
MES order number S30002090200 Deburring	Article SAR-VPO011-M-9005 Quantities (target/yield/scrap) 1200 / 660 / 0

Staff logged on		
Person 10002630 Name White, Susan	Person 85741 Name Meyers, Joseph	Person 40563 Name Green, Mary

At the bottom left, there is a 'PZE' button. The bottom right corner shows the date and time: 23.11.15 16:22:40.

The basic display shows the third list "Input materials currently logged on" for machines for which the "batch management" option is configured. All active input batches of the selected machine are displayed in this list.

OP logon with input batches

A workflow including two tabs is opened by clicking the button highlighted with the red frame for logging on operations. The operation to be logged on is selected in the first tab "select operation".

The "log on operation" tab is reached by clicking "Next" where in addition to the selected OP, the defined material components are displayed in a list.

By entering a batch number in the "input batch" field and clicking the "report batch" function a batch may be logged on as input material for a component. During the entry process, the terminal checks whether or not the batch number is known to the system and may be logged on. This is also described in detail in the document dealing with the [Input batch change](#).

"Batch" field

When an OP is logged on, a batch number is created simultaneously for the next output batch to be produced. The batch number may be assigned automatically or manually (please also see the settings of "workplace configuration"). The generated batch is created with the batch number in the system and set to the "running" status.

Provided that all required input materials are logged on, the OP may be started via the "OK" button in the "OP logon" dialog. Whether input material has to be logged on or not, can be defined in the assigned material type of the component.

Once the OP has been logged on successfully, all active input batches of the selected machine are displayed in the material list.

If batches are logged on along with the OP and the user cancels the process or cannot log on the OP due to validation checking, the input batches will be logged off automatically for this OP. In this case, batches are always logged off without indicating the consumption quantity. By way of the following warning message, the user may confirm the logoff process:

The function logging off input batches automatically can be activated/deactivated by an option in the hyaipcfg.ini file:

HYAIPCFG.INI

<pre>[MPL-Options 0/2xxx] ForceAutoLogOffInputBatches=0</pre>

Logon of unplanned input material

In addition to planned materials, it is also possible to log on "unplanned" material for an OP, using an additional feature for the OP. If the "replaceable" option is set to "J", the user is able to assign the respective component manually, when batches are logged on. However, the logon is only allowed if the material type of the input batch matches that of the component.

In the selection list the components are filtered by the material number and displayed as follows for selection:

Logon of unknown batches

An input batch, which is not yet known to the system, may be logged on for an OP using the "create unknown batches" option in the basic parameter settings.

In this case, when input batches are logged on, the system searches for a valid assignment of input material to the material type of the selected component. Provided that a matching assignment is found and the "allow entry of unknown input batches" option is configured for the material type in the "input batch processing" tab, the batch is generated by logging it on to the system. Then it is set to the "running" status. The batch is initially created in a quantity of 1.000.000.000.

Logoff/interruption of OPs

A running OP may be interrupted or logged off by clicking the "logoff/interrupt OP" button. Then a dialog opens, where the following selection can be made:

If "log off OP" is clicked the logoff dialog opens that contains the same input fields like the [Output batch change](#) dialog.

Thus, the output batch that is currently active is completed, when OPs are interrupted or logged off.

9 Input Batch Change

Summary

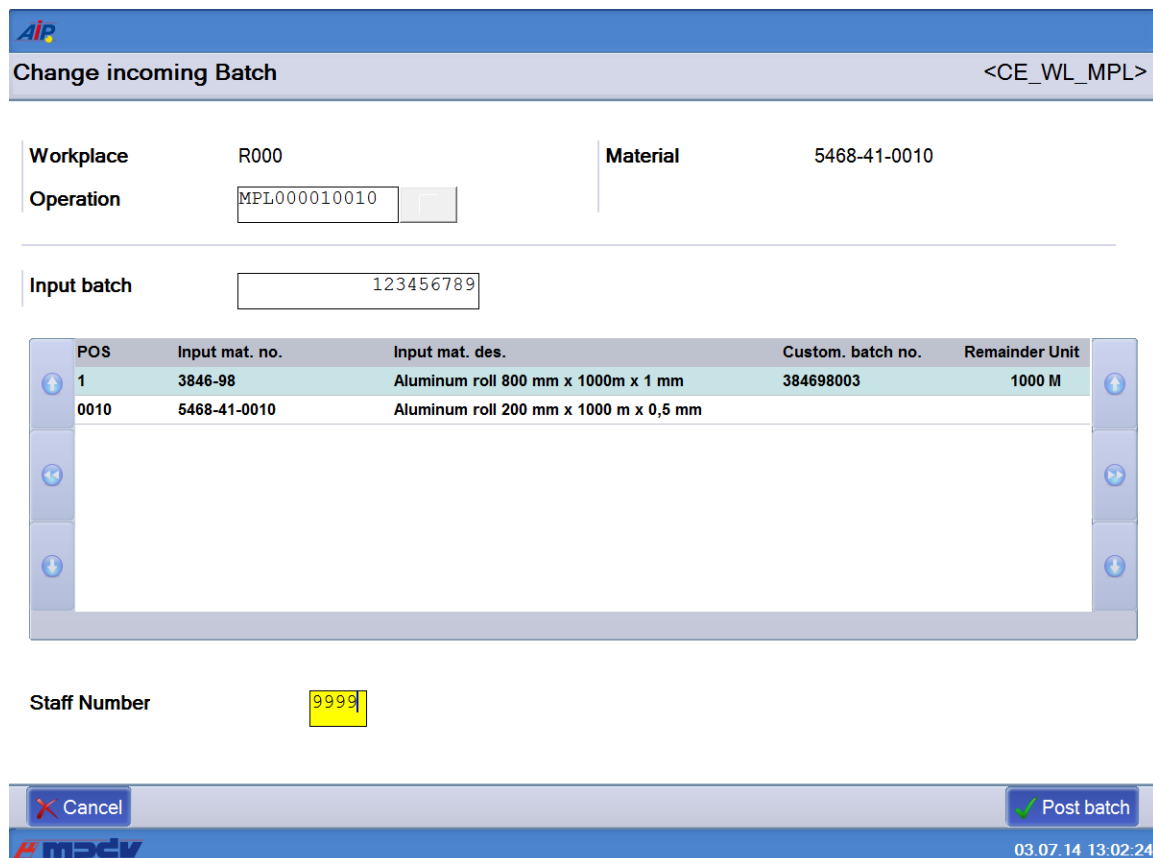
Input material and/or relevant input batches can be changed for a running OP if the "input batch change" option is clicked.

Configuration

Further system configurations are not required to be able to use the input batch change dialog.

Dialog

Basic screen



Change incoming Batch <CE_WL_MPL>

Workplace: R000 Material: 5468-41-0010

Operation: MPL000010010

Input batch: 123456789

POS	Input mat. no.	Input mat. des.	Custom. batch no.	Remainder Unit
1	3846-98	Aluminum roll 800 mm x 1000m x 1 mm	384698003	1000 M
0010	5468-41-0010	Aluminum roll 200 mm x 1000 m x 0,5 mm		

Staff Number: 9999

Cancel Post batch

AIP 03.07.14 13:02:24

Log input batch off:

Input batches may be changed by entering a currently active batch number or by entering a new batch number. When logging batches off, it is also possible to enter the status and consumption of the batch to be logged off.


<CE_AB>

Unregister Incoming Batch

Batch	384698003		
Material	3846-98		
BOM item	1		

Status
☐ PROCESSED
☐ LOCKED
☒ with remaining qty.

Consumption 10 M

Comment

Staff Number

Unregister Incoming Batch


03.07.14 13:03:20

Options when logging input batches off:

F1 - PROCESSED

The batch is set to the "processed" status and the remaining quantity that is still available is set to 0. A consumption posting is generated as goods issue for the current, remaining quantity.

F2 - BLOCKED

The batch is set to the "blocked" status. A consumption entered additionally is deducted from the current, remaining quantity as goods issue.

F3 - with remaining quantity

The batch is set to the "free" status. A consumption entered additionally is deducted from the current, remaining quantity as goods issue. If the remaining quantity that is still available becomes ≤ 0 , the batch status automatically switches to "processed".

Consumption

The entered consumption (unit of the input material) is deducted from the remainder of the batch and a goods movement is generated.

Comment on batch

The comment entered is saved as information for the batch.

Log input batch on:

Provided that the batch is known, batch data is displayed in an intermediate dialog where the logon may be confirmed.

Provided that the batch could be logged on, it is taken over to the material list in "customer batch number" and thus the change is completed.

However, in case the logon is inadmissible as the input material does not correspond to that of the component, the logon is rejected by an error message.

10 Output Batch Change

Summary

Output material may be changed for a running OP using the "output batch change" option.

Configuration

Further system configurations are not required to be able to use the output batch change dialog.

Dialog

Change Outgoing Batch
<CA_WL_MPL>

Workplace
R000
Material
5468-41-0010

Operation
MPL000010010

New batch
PR4RH6A112

Target buffer
MP_N_50505

Transp. unit
ROLLCONT

Quality
Yield
Scrap

Qty
0
M

Reason
0

Comment

Input mat. no.	Input mat. des.	Batch No.	Custom. batch no.	Remainder Unit
3846-98	Aluminum roll 800 mm x 1000m x 1 mm			0 M
5468-41-0010	Aluminum roll 200 mm x 1000 m x 0,5 mm			

Staff Number

Cancel
Preced. batches
Change i.batch
Change Outgoing Batch

03.07.14 13:09:57

The input batches that are logged on are displayed with their available remaining quantity in a list within the output batch change dialog. The following data may be entered:

Target buffer

Material buffer for which the current batch is to be produced. The output material buffer of the machine is pre-assigned as default value.

Transport unit

A transport unit defined within the system may be assigned here. The selection refers to transport units that were assigned to the material type of the OP.

Comment on batch

In this field a comment may be saved for the batch to be produced.

Quantity

The batch to be produced is posted with the quantity entered here. The quantity is taken over as primary quantity of entry to the order and machine and a goods movement is generated as goods receipt.

Quality

A batch may be classified as yield or scrap quantity. The system posts yield batches with the "free" status. Scrap batches automatically get the "blocked" status. When scrap is selected, a valid scrap reason has to be assigned.

"Preceding batches" function key:

This button opens a list with output batches which have already been produced for this OP.


Display preceding batches
<C_VLOS_MPL>

Workplace

R000

Operation

MPL000010010

Preced. batches

Input mat. no.	Input mat. des.	Custom. batch no.	Target
5468-41-0010	Aluminum roll 200 mm x 1000 m x 0,5 mm	PR4RD97116	MP_N_50500
5468-41-0010	Aluminum roll 200 mm x 1000 m x 0,5 mm	PR4RD97115	MP_N_50500
5468-41-0010	Aluminum roll 200 mm x 1000 m x 0,5 mm	PR4RD97112	MP_N_50500
5468-41-0010	Aluminum roll 200 mm x 1000 m x 0,5 mm	PR4RD96116	MP_N_50500
5468-41-0010	Aluminum roll 200 mm x 1000 m x 0,5 mm	PR4RD96115	MP_N_50500
5468-41-0010	Aluminum roll 200 mm x 1000 m x 0,5 mm	PR4RD96114	MP_N_50500










 Cancel
Re-print


03.07.14 13:10:35

"Change inp. batch" function key:

Using this button the user can switch to the "input batch change" function.

New batch

When a current output batch is completed, a new batch number is simultaneously created for the next batch. The batch number may be assigned automatically or manually. The batch generated in this way is created with the batch number in the system and set to the "running" status.

11 Batch Information

Batch information

Batch information is displayed in a dialog, when the “batch info” icon is clicked.


Dialog open for more than 5min

<C_LOS_INFO>

Batch information

Batch	SAM101
Material	453-9900
Designation	Rahmen 453-9900
Type	RAHMEN
Class	G
Operation	
	SNRAUNR90010
Machine	R000
Material buffer	
	MP_V_50505
Remaining qty.	0,0 ST
Prod.date	27.06.2014 08:36:19
Status	L
Note	

OK


CTAIP V2.0.3.4 Server.scc7:4 TNR:131

03.07.14 11:03:54

12 Entry of Batch Attributes

Summary

In case batch attributes are defined to be recorded at the AIP terminal for the material type of the running OP, another input dialog is opened, when the output batch is changed. The dialog opens additionally after clicking "OK" in the output batch change function, interrupt OP and finish OP function.

Using batch attributes, numeric and alphanumeric values may be recorded which are then saved for the produced output batch in an additional table.

Configuration

By configuration it is possible to record any number of additionally required batch attributes for the material type.

Dialog

Example when two additional batch attributes are collected for an output batch:

Length (CM) :	0.00	✓ OK
Width (CM) :	0.00	✗ Cancel
Type () :		

13 Enter Goods Receipt Batch

Summary

A new goods receipt batch may be created in the system via the "enter GR batch" button.

Configuration

Further system configurations are not required to enter goods receipt batches.

Dialog

Wareneingangslos erfassen
<C_GEN>

Los

Arbeitsplatz

Arbeitsgang

Material

Zielpuffer

Transporteinheit

Loshinweis

Qualität

☒ Gutmenge

☐ Ausschuss

Menge

Grund

Personalkartennummer

Abbrechen

WE-Los erfassen

Having clicked the "OK" button, the batch is created and the dialog remains open for further entries.

Batch numbers may be generated automatically or manually depending on the configuration. Moreover, the following data is saved at the batch.

Workplace

Machine where the batch was recorded

Operation

AIP-TRT_82.docx

Version: 1.1.23049

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Order where the batch was recorded

Material

Material number of the batch

Quantity and unit

The batch is created with the quantity and unit entered here. A goods receipt is posted with the quantity.

Quality

A batch may be classified as yield or scrap quantity, rework or open quantity. The system creates yield batches with the "free" status. Scrap batches automatically get the "blocked" status and the batch class "scrap". When scrap is selected, a valid scrap reason has to be assigned additionally with respect to the corresponding workplace.

Target buffer

Material buffer for which the current batch is to be produced. The output material buffer of the machine is pre-assigned as default value.

Transport unit

A transport unit defined within the system may be assigned here.

Comment on batch

In this field a comment may be saved for the goods receipt batch.

14 Repost Batch

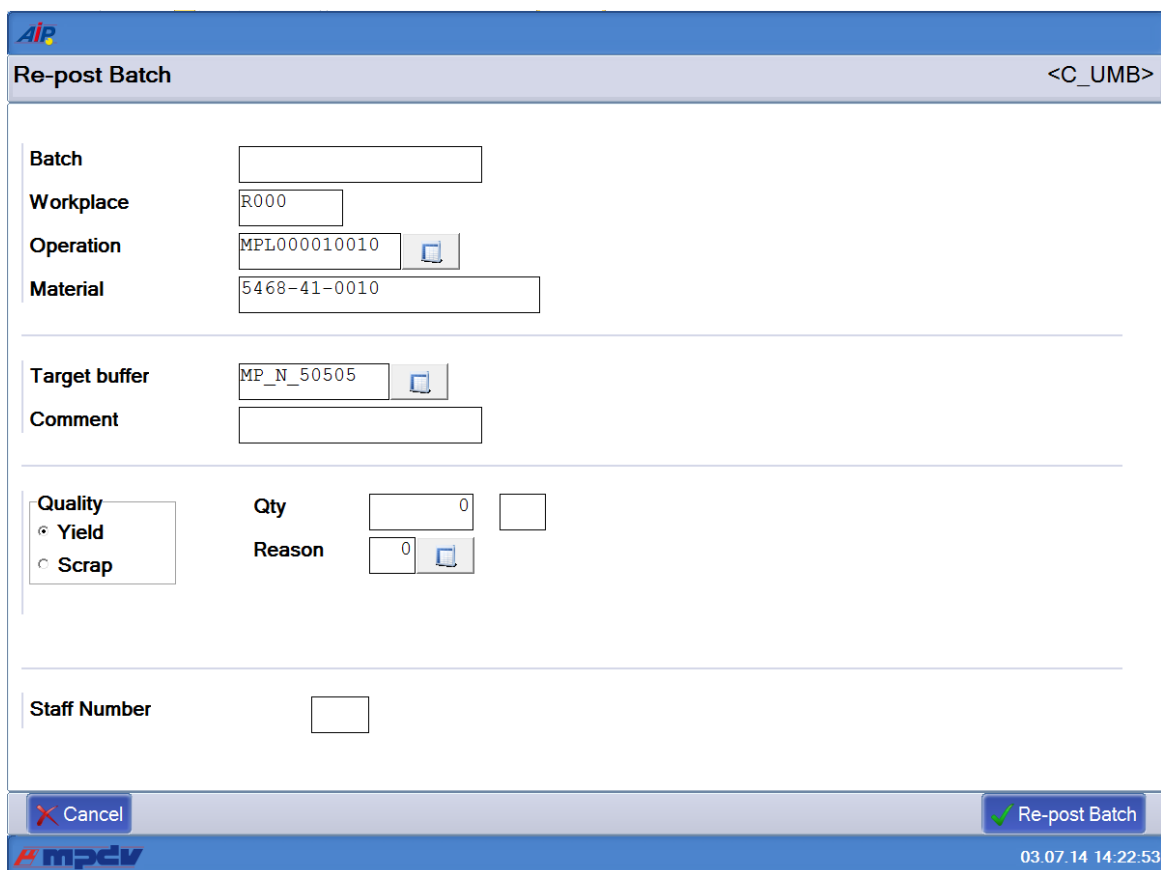
Summary

Using the "repost batch" button, an existing batch may be reposted to another material buffer.

Configuration

Further system configurations are not required to repost batches.

Dialog



The dialog box is titled "Re-post Batch" and has a "<C_UMB>" button in the top right corner. It contains the following fields and controls:

- Batch**: A text input field.
- Workplace**: A text input field containing "R000".
- Operation**: A text input field containing "MPL000010010" with a small icon to its right.
- Material**: A text input field containing "5468-41-0010".
- Target buffer**: A text input field containing "MP_N_50505" with a small icon to its right.
- Comment**: A text input field.
- Quality**: A group box containing two radio buttons: "Yield" (selected) and "Scrap".
- Qty**: A text input field containing "0" with a small icon to its right.
- Reason**: A text input field containing "0" with a small icon to its right.
- Staff Number**: A text input field.

At the bottom of the dialog, there are two buttons: "Cancel" (with a red X icon) and "Re-post Batch" (with a green checkmark icon). The bottom status bar shows the "mpcv" logo and the date/time "03.07.14 14:22:53".

Having clicked the "OK" button, the batch is reposted to a new material buffer and the dialog remains open for further entries.

As an alternative, the batch can also be reposted from yield to scrap.

15 Display of "Produced Batches" in AIP

Overview

Using the [3rd list of the machine master](#), it is possible to display the output batches produced for a running operation that is subject to batch management.

A default number of 20 output batches is displayed for each machine in the list. Output batches (yield and scrap batches) are shown, which have been produced at this terminal since output batches were changed.

As this list is only kept locally by the terminal, it is not synchronized with the server, when AIP is started.

Configuration

The settings required to use the third list can be found in the document dealing with [configuration](#).

Display

Produced output batches				
...	Batch HU4RD9711P Article 5468-41 Aluminum roll 200 mm x 1000 m	Batch HU4RD9711O Article 5468-41 Aluminum roll 200 mm x 1000 m	Batch HU5AD1G117 Article 565-941 Aluminum roll 100 mm x 1000 m	Batch HU4RD9711Q Article 5468-41 Aluminum roll

If several third lists are enabled, you can switch between the lists in the header line.

The list shows, among others, the article, article name and batch number. Some of the fields might not be completed depending on the data collection scenario when generating output batches.

16 Settings for List of Produced Output Batches

Activation at the machine

For the purpose of activating the list of produced output batches in the basic screen of the terminal, the display of the third list of the machine to be activated has to be activated in the [machine master record](#) (Workplace configuration → Entry → Display 3rd list).

Definition of the Number of Entries in the List

The number of entries in the list may be customized by extending the data provision for the machine list (mnr.lst) at the terminal on the server side (extended customizing). For this purpose, the additional column "MNR.NUMBER_OF_BATCHES" has to be provided at the server.

Customization of Visible Columns in the List

The list contents can be configured in section [MNR_AMAT.LST] of the file ctaiply.ini.

Example:

CTAIPLAY.INI
<pre>[MNR_AMAT.LST] GRID_FONT=Arial GRID_FONTSIZE=9 GRID_COLOR=clBlack GRID_BACKGROUND=clWhite GRID_ORDER=ROW.IDX=- GRID_CAPTION=Produzierte Ausgangslose EXAMINE_SCANEXPR1=KLASSE=G EXAMINE_SCANCOLOR1=clGreen EXAMINE_SCANEXPR2=KLASSE=A EXAMINE_SCANCOLOR2=clRed ; ROW.IDX=N10,50,R,Row CNR=C20,150,L,Losnummer ; CNR=*CNR,Los ATK=C25,125,L,Artikel ; ATK=*ATK ; KLASSE=C3,40,Z,* MENGE=N12.0,70,R,Menge EINH=C3,30,Z,ME DAT=dd.mm.yyyy,70,L,Datum ZEI=hh:mm:ss,60,L,Zeit ATKBEZ=C30,200,L,Artikelbezeichnung</pre>

Configuration of Server-Based Comparison

The server-based comparison is activated by an entry in the customer-specific configuration file "ctwinlisten.ini". As with [hytnrcfg.ini](#), this file can be maintained both globally (for all terminals), on a terminal group level, and on terminal level.

Activation is effected by entering a LOADCYCLE larger than 0 (seconds).

CTLISTEN.INI
[#LIST#TNR-ALOSE] LOADCYCLE=900

The prerequisite for the server comparison is the default configuration file "ctlisten.cfg", which contains the application-specific configurations of the server list.

CTLISTEN.CFG
[#LIST#TNR-ALOSE] CMD=DLG=LIST;13 MOD=P ; (Default <ANZ=250>) LOADCYCLE=0 QUEUEEMPTY=TRUE FORCENOTIFY=TRUE

17 Advance Logon of Input Batches

Summary

The process might require an input batch to be logged on in advance and set up accordingly on a machine, while the preceding input batch is still being used for a material.

This situation frequently occurs at very large machines processing, for example, roles or belts that are uncoiled as input batch at the beginning of the machine and coiled up as output batch at the end of the machine.

As the users are mostly busy with activities at the end of the machine at the time when the input batch actually needs to be changed, they cannot perform the input batch change and, as a result, they are provided with the opportunity to log the next input batch on already in advance for an order/OP.

Then the input batch can actually be changed by logging a new OP on or a project-specific call can be established.

General / usage

The function for logging input batches on in advance is used to be able to "set up" and "log on" the next input batch while an OP and input batch are still running. This next input batch is not yet running but assigned the "logged on in advance" flag.

An input batch may be logged on in advance for a currently running OP or a prepared OP.

Configuration

The settings required for using the function "advance logon of input batches" is described [here](#).

Procedure

The procedure for using the function "advance logon of input batches" or the logical process is described [here](#).

Dialog

Basic screen

The basic AIP screen shows the function key "Advance logon of input batch" (preregistration of input batch). The dialog for logging input batches on in advance may be used by clicking this function key.

Machines/workplace:

Machine/	Workplace	Status	Status since	Note
CPL	Continue Pickling Line	PRODUCTION	26.03.13 08:00	
CRM	Cold Rolling Mill	PRODUCTION	26.03.13 08:18	
CEL	Electrolytic Cleaning Line	NO JOB	26.03.13 08:00	
CAF	BAF	NO JOB	26.03.13 08:00	

Page 1 • Page 2

Lock prod.status Modify status Log on operation

Operations on workplace:

Article	Order	Operation	Target qty.	Yield	Scrap	N	T
---------	-------	-----------	-------------	-------	-------	---	---

Comment Modify target qty. Inp. batch change **Preregistration of input batch**

Registered input materials

Position	Batch	Material	Material designation	Remaining
01	R835-022	A06A2100926	2.10MM X 926MM HR COIL - SAE1006(BS	18 mt

CTAIP V2.0.2.67 Server:win2008-10:3 TNR:999 26.03.13 12:23:25

Advance logon of input batches (CE_VWL_MPL)

The user selects the workplace to which an input batch is to be logged on in advance in the basic screen. The below dialog (CE_VWL_MPL) opens by clicking the function key "Advance logon of input batches" (preregistration of input batch).

Preregistration of input batch <CE_VWL_MPL>

Workplace CRM Material CCSD1901223

Operation 22999998010

Input batch

POS Input mat. no.	Input mat. des.	Cust. Batch nr.	Remainde Unit
01	A06A2100926	R835-022	18 mt

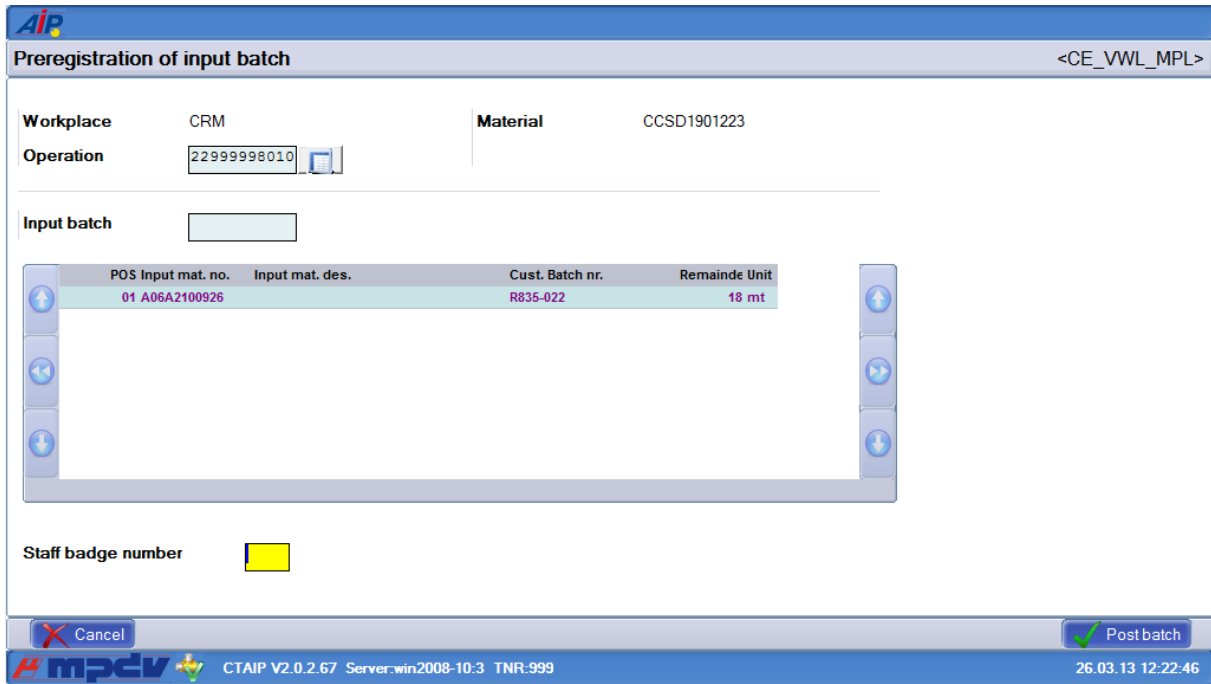
Staff badge number

Cancel Post batch

CTAIP V2.0.2.67 Server:win2008-10:3 TNR:999 26.03.13 12:22:46

If an operation is currently running/logged on to the workplace, this one will be selected by default. The input batch (that is to be logged on in advance) is entered/scanned for the selected BOM item. Advance logon of input batches is started by clicking the button "post batch".

At first the input batch is checked for validity (dialog CE_VAN). The material number of the input batch is checked against the material number of the component list or the BOM item. The input batch is logged on in advance, once the button “log input batch on in advance” has been clicked:



POS Input mat. no.	Input mat. des.	Cust. Batch nr.	Remainde Unit
01 A06A2100926		R835-022	18 mt

Finally, the input batch that has been logged on in advance is displayed in purple in the BOM of the component.

The dialog can be closed with the “cancel” key.

18 Configuration: Advance Logon of Input Batches

Configuration: display of third list on the AIP (INI configuration)

Set the parameters/values below in the INI configuration to display the input batches logged on in advance in the third list on the AIP:

- INI name: MPL
- Section: MPL_VANCNR
- Key/Value: TNR_VANCNR =Y

INI configuration	
Section	MPL_VANCNR
Key	TNR_VANCNR
Value	Y
	☒ Active
Commentary	Display preregistered batches in the 3rd list



When you have made the settings in the INI configuration, restart the terminal. The INI configuration is only activated when the terminal is restarted.

Configuration: logon of input batches with OP logon on the AIP (INI configuration)

Set the parameters/values below in the INI configuration to log on the input batches logged on in advance at the same time with an OP on the AIP.

- INI name: MPL
- Section: MPL_VANCNR
- Key/Value: USE_VANCNR =Y

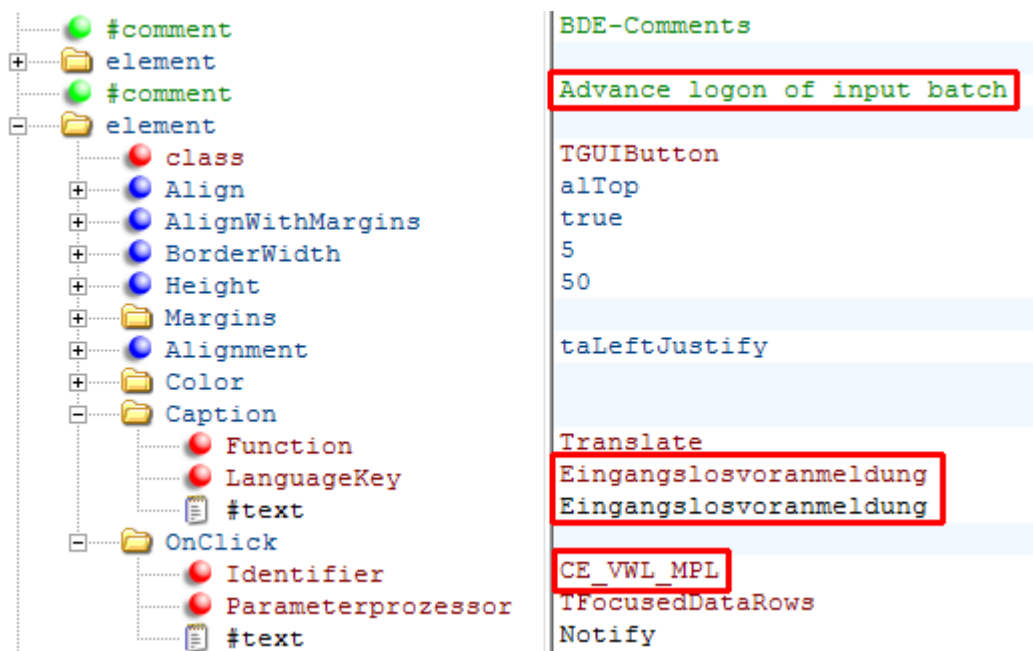
INI configuration	
Section	MPL_VANCNR
Key	USE_VANCNR
Value	Y
	☒ Active
Commentary	Use preregistered batches when logging an operation



When you have made the settings in the INI configuration, restart the terminal. The INI configuration is only activated when the terminal is restarted.

Configuration of keyboard layout on AIP2

To define the function key "Advance logon of input batches", copy the layout *guiV_anr.xml* to *guiV_anr_In.xml* and copy the key "BDE comments". Change the following entries in the configuration of the copied key:



If you do not use the AIP2 tile view, define the function key in the file *ctaipbut.ini*:

```
[ANR-LN-Page2]
```

```
1=A_INFO.Dialog1,L,BDE-Kommentar,Attach Notes.png
```

```
2=A_SMG,L,Sollmenge ändern,Shipping Box Open Move Down Up.png
```

```
3=A_ELW,R,Eingangsloswechsel,CE_WL.png
```

```
4=CE_VWL_MPL,R,Eingangslosvoranmeldung,CE_WL.png
```

```
5=%BART:CAQ=J%CAQ_DC_T,R,Prüfung durchführen,Generators.png
```


Configuration: Highlighting the input batches logged on in advance on the AIP (ctaiplay.ini)

To highlight the input batches logged on in advance in color in the list of input batches, material list and BOM on the AIP, make the following entry in the configuration file "ctaiplay.ini". Store the file on the server.

Entry in Ctaiplay.ini:

Sections: [list of input batches], [material list] and [PRT list (KOMBI)].

...

```
EXAMINE_SCANEXPR1=CST=X
EXAMINE_SCANCOLOR1=C1Purple
```

...

Make sure that each key is only available once in a section.

If you want to store several rules, you can integrate this using sequence numbers at the end of the keys:



```
EXAMINE_SCANEXPR1=ART=T|Z
EXAMINE_SCANCOLOR1=c1Blue
EXAMINE_SCANEXPR2=ATKDIFF=F
EXAMINE_SCANCOLOR2=c1Green
EXAMINE_SCANEXPR3=ATKDIFF=J
EXAMINE_SCANCOLOR3=c1Red
```

19 Throughput Batch Processing

Overview

Throughput batch processing is a special application case in material and production logistics. In this case, an output batch is to adopt the visible batch number of the input batch used.

Usage/Procedure

In contrast to normal input and output batch processing in an operation requiring batch management, in throughput batch processing an input batch with a batch number is used and this batch number is handed down to the output batch.

Throughput batch processing is used for instance if the condition of a batch/material might change after a work stage but the external batch number (e.g. on a label) is to be retained.

As a consequence, no new material will occur when the output batch is changed and the batch number can even be handed down through several process steps.



The operator uses an input batch on a machine/operation. The output batch number is not changed and hence remains identical to the input batch number.

The so-called throughput batch number (throughput batch number/external batch number) therefore remains identical. Within the system, however, a unique batch number (HYDRA batch number/internal batch number) is still used and/or generated for each production level/after each output batch change, since every object within the system is unique.

If, for instance, a selection according to throughput batch numbers is made in the batch data overview, several entries with different internal batch numbers will be obtained for each throughput batch number; these internal batch numbers ensure unambiguity within the system and consequently allow for a historic observation of the "throughput batch".

This means that a total of three different statuses are considered for the process description:

- Status 1: The batch as an input batch (prior to logon)
- Status 2: The batch status on the machine (running on OP)
- Status 3: The batch as an output batch (after logoff)

The batch as an input batch (prior to logon):

Prior to logging on the batch, the information on the batch (e.g. in the batch data overview) reads as follows:

- the throughput batch number *dllsnr* "**DLLOS01**" is identical to the batch number *losnr* "**DLLOS01**"
- the batch status is "**F**" (free)
- the throughput batch flag *dll_kennz* is "**N**"

The batch status on the machine (running on OP)

When an operation is logged on to the terminal, the material (the material number) is used according to the component list from the OP in order to determine whether it is treated as a throughput batch on the basis of the material type entered for this material. This procedure also enables logging on the throughput batch number on this operation. It is not possible to log on more than one input batch/material as "throughput batches".

After logging on the batch (with throughput batch number/external batch number) on the operation/machine, the information on the batch reads as follows:

- the batch status is changed to "**L**" (running)
- the batch receives the throughput batch flag *dll_kennz* "**E**" (throughput input batch running)

After logging off the output batch on the OP/machine, a new internal object is created in the system to take over the throughput batch number of the input batch.

The information on the new output batch then reads as follows:

- a new batch with *losnr* **PR41E9C114** and *dllosnr* **DLLOS01** is created in the output buffer
- the status of the new batch is changed to "L" and
- the batch receives the throughput batch flag "G".
- The quantity on the new batch is identical to the quantity of the original input batch.

Functionality: Console Evaluation of Throughput Batches

An operator can identify and trace a created batch via the external batch number (e.g. in order to be able to forecast when a specific batch will leave production).

Material Movement Functionality

An operator can call up the material movements (goods issue/goods receipt) for an input batch/output batch.

Batch History Functionality

An operator can list the batch history of an output product in order to be able to trace the manufacturing process of a batch for analyses.

Batch Tracing Functionality

An operator can use the batch tracing functionality to verify through which machines/operations the throughput batch was produced. The operator thus sees the material's route through production.

20 Throughput Batch Processing

Summary

A machine/workplace may also be configured in "throughput batch mode".

In [throughput batch mode](#) input material is continued being processed with unchanged number (throughput batch number) using an OP.

Please note: At machines with "throughput batch mode" it is impossible to log operations on at the same time.

Configuration

How to configure throughput batch processing is described [here](#).

Dialog

The entry functions for throughput batch processing at the terminal are identical to those for active batch tracing at the machine.

21 Configuration for Throughput Batch Processing

Overview

Usage

In material and production logistics, the batch number of an input batch can be handed down to an output batch and thus make so-called throughput batch processing possible.

However, in order to enable throughput batch processing, the following configurations are required on various objects.

Machine Configuration

In the machine configuration (Master data → Workplaces/machines → Workplace configuration), the batch management in the Workplace configuration (MPL) tab is to be set to value "D".

When throughput batch processing is active on the machine, the "Automatic generation of batch number" option cannot be used, since in this case the batch number will always be handed down by the input batch.



The entry of machine cycles in combination with throughput batch processing is not used in general, since there is always a 1:1 transfer of input batches into output batches. For this reason, throughput batch recording is generally only used in connection with manual unit posting at the terminal (e.g. use in furnace, conditioning, etc.).

Configuration of Material Type

In the master data configuration (Master data → Material → Material type), the [material type](#) is to be configured in such a manner that the batch number is "handed down" and the input batch is only valid for one output batch.



- Retrograde inventory management is generally not performed for the component whose batch number is transferred as a throughput batch, since consumption is always 1:1.
- A parallel log-on of the input batch on several machines is not supported by the system.
- The entry of unknown input batches is not supported by the system.

Configuration at the Operation

The configured material type is to be entered as the material type at the operation.

The operation is to be identified as requiring batch management.

Configuration at the Operation - Component

The configured material type is to be selected as the material type at the component.



In addition to the component for which the batch number is to be handed down, other material components can be maintained at the operation. These continue to be taken into account in the usual way in the course of batch log-on and consumption recording.

22 Batch Consumption

Usage

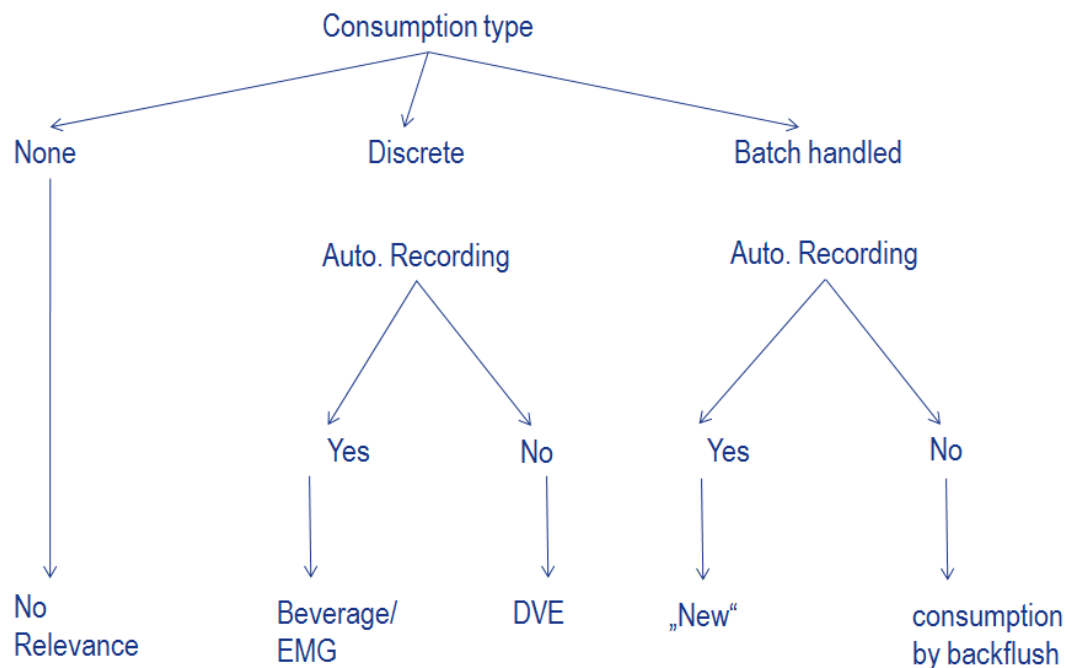
Material is used and represented in the system by:

- material postings to regulate inventories with/without ERP and without tracing
- batch-related material postings to trace back the recorded parts/materials

Subject to the type in use, consumption can be recorded differently in the system.

Procedure

These types of consumption recording are used in MES:



Discrete consumption recording:

Discrete consumption recording is used every time when a discrete amount of consumption can be entered for the used components by the user or a counter. Data is only entered for the component/material number. Data may be collected:

- automatically (configuration of a consumption meter per material)
- manually (the user enters material consumption manually)

Batch-related consumption recording:

Batch-related consumption recording is used every time when batches are used for the components for which a consumed quantity can be entered by the user or a meter. Data is collected regarding the input batch. Data may be collected:

- manually (dialog to log off the input batch and to enter batch consumption manually)
- in a retrograde manner/backflush (automatic calculation of batch consumption by generating the output batch quantity)
- automatically (automatic collection of batch consumption by a consumption meter)

Discrete consumption - manual



How to enter discrete consumption is described [here](#).

Discrete consumption - automatic

General

Automatically recorded consumption is collected by a meter configured at the machine. Data is collected for a material type of materials included in the component list.

Configuration

These configurations have to be set in the system if material consumed discretely is to be indicated/counted by a meter:

- Component of the OP:
The "consumption type" has to be set to "D = discrete".
- Material type of the material:
The option "inventory management" has to be set to "N = No".
- Meter for the material type of the material:
Configure meter like MDE meters.
Option "compensation with material" = yes
Option "material type" = material type of the material that is consumed

Posting/result

A goods issue is generated for automatically recorded consumption in the system.

Manual batch consumption

General

Manual batch consumption is entered by the input batch change function. The user enters the consumed quantity when logging the used input batch off.

Configuration

These configurations have to be set in the system if material is to be consumed manually as input batch:

- Component of the OP:
The option "consumption type" has to be set to "L = Backflush/with batch reference (retrograde)".
- Material type of the material:
The option "inventory management" has to be set to "E = Yes, when logging input batch off".

Posting/result

The consumed quantity is deducted from the remaining quantity of the input batch and the batch shows the reduced "remaining quantity" and the initial quantity.

A goods issue is generated for consumption in the system.



How to enter batch-related consumption is described here.

Retrograde batch consumption

General

Retrograde batch consumption is calculated continuously as the output batch quantity increases. When logging the input batch off, the remaining quantity of the input batch is reduced by the calculated consumption quantity. Usually, the user does no longer enter a quantity when logging the used input batch off.

Configuration

These configurations have to be set in the system if material is to be consumed in a retrograde manner as input batch:

- Component of the OP:
The option "consumption type" has to be set to "L = Backflush/with batch reference (retrograde)".
- Material type of the material:
The option "inventory management" has to be set to "R = Yes, backflush (retrograde)" or "G = Yes, backflush (only with YIELD batch), retrograde".

Posting/result

The quantity calculated in a retrograde manner is deducted from the remaining quantity of the input batch and then the batch shows the reduced "remaining quantity" and the initial quantity.

A goods issue is generated for consumption in the system.

Automatic batch consumption

General

The automatically recorded batch consumption is collected continuously as the meter quantity increases. When logging the input batch off, the remaining quantity of the input batch is reduced by the automatically recorded consumption quantity. Usually, the user does no longer enter a quantity when logging the used input batch off.

Configuration

- Component of the OP:
The option "consumption type" has to be set to "L = Backflush/with batch reference (retrograde)".
- Material type of the material:
The option "inventory management" has to be set to "R = yes, backflush (retrograde)".
- Meter for the material type/BOM item of the affected material:
Configure meter like MDE meters.
Option "compensation with material" = yes
Option "material type" = material type of the material that is consumed
or
Option BOM item = BOM item of the material that is consumed



If the BOM item is used within meter configuration, it is important that within the OP's component list the material is always used as the same BOM item (from ERP work plan).

Posting/result

The automatically recorded quantity is deducted from the remaining quantity of the input batch and then the batch shows the reduced "remaining quantity" and the initial quantity.

A goods issue is generated for consumption in the system.

23 Discrete Consumption Input

Usage

Key input values when collecting shop floor data are times and quantities. While times or durations are used to describe the time effort that was required to manufacture a material, quantities document the entire scope of the produced material. The objective is to process or output the material quantities defined in the order or the operation

In most cases, the quantities entered are sufficient to be able to execute the actions that will change stock quantities in the upper-level ERP system accordingly:

- Goods receipt from production

The finished quantity will increase stock in the ERP system.

- Goods issue from production

Based on the finished quantity, the consumption of the material that flowed in (so-called input material) can be calculated in reverse order of the production process in the ERP system, accounting for the bill of materials the order is based on. This will lead to a reduction of stock in the ERP system.

However, this kind of consumption calculation is oftentimes not enough to ensure a "clean" inventory management in the upper-level ERP system. Instead, there is a need to discretely enter order-related material consumption and to then post the consumption later in the ERP system.

You make use of this functions package if you would like to enter material consumption discretely and without a batch reference and upload it to an inventory management system.

Integration

The function can be integrated into a system dedicated to shop floor data collection (BDE), or also be used within the context of material and production logistics (MPL/ TRT).

These functions are used:

- Functions for discretely entering order-related material consumptions without a batch reference.
- Posting dialog at the Windows Terminal AIP to input discrete consumptions relating to material components during operation logoff, OP interruption or partial confirmation/upload.
- Entry based on the produced and manually entered total quantity or yield and/ or scrap.

- Providing material consumptions for confirmation/upload from HYDRA to the inventory management system in HYDRA standard format (requires that the interface used to upload material and batch data is licensed and activated).

24 Discrete Consumption Input at AIP

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Summary

The function described below makes it possible to enter material consumption at the AIP shop floor terminal so that it can be uploaded via an interface in the form of goods movements.

Material consumption can be entered at the AIP shop floor terminal while an operation is being interrupted, while an operation is being logged off or when entering a partial confirmation/upload

Configuration

The document dealing with the [configuration](#) is to be taken into account for discrete consumption input.

Dialog and procedure

AIP provides the following features to enter material consumption based on the produced (total) quantity.

In the dialogs

- Log operation off (A_AB)
- Interrupt operation (A_UN)
- Partial confirmation/upload for operation (A_TR)

an additional "consumption" button is available with which the "Component consumption posting" dialog can be called up.



It is also possible to make the button available that is used to call up the "Component consumption posting" dialog from the MPL specific dialogs used for operation interruption (A_UN_MPL, A_UN_RF, A_UN_RS) or for operation log off (A_AB_MPL, A_AB_RF, A_AB_RS). However, what needs to be considered in this regard is that in MPL, consumption of batch-related material is posted differently.

Figure: "Partial confirmation/upload (A_TR)" dialog with "Consumption" button in AIP

The input dialog opens after the "Consumption" button is pressed.

The status information shown includes the workplace displayed in the dialog from which the function is called up, the operation and also the yield and scrap quantities entered in the dialog from which the function is called up. When calling up from an MPL dialog, the entered quantity is displayed based on whether classified ("Quality") as scrap or as yield.

Furthermore, a table is displayed showing the components that are flagged in the component list as consumption type "D". The following data is displayed in the table.

BOM item

BOM item

Input.Mat.No.

Material number of the material component.

Input.Mat.Des.

Material designation of the material component.

Consumption

Calculated consumption based on the quantity entered in the dialog from which the function is called up:

$$\text{Rechnerischer Verbrauch} = (\text{Gutmenge} + \text{Ausschuss}) * \text{Einsatzmenge}$$



Only yield and scrap quantities are taken into account when calculating the consumption.

Quantities are not set off against each other (e.g. scrap set off against yield) when the calculated consumption is determined.

Unit

Quantity unit (unit of the input quantity).

Input quantity

Input quantity required to produce one quantity unit of the output material.

When a component is selected from the list, the calculated consumption is proposed in the "Consumption" input field. If the actual consumption deviates from the calculated consumption, the operator can now modify this quantity; it is transferred to the list. Any consumption that was modified manually is shown highlighted in "green" in the list (e.g. 

By pressing the "Reset" key, the component list can be called up again; this will recalculate the consumption quantities; any consumption quantities that were already modified will be overwritten.

After confirming this dialog and the dialog from which the function is called up, the consumption quantities are updated and are thus transferred to the server via the posting. There, the consumption is updated as the status in the component list at the operation. In addition, one material movement (goods issue) is written for each component.



The material consumption is only posted as a goods movement if the dialog was called up explicitly. Material consumptions that were not posted are not automatically posted when the operation is logged off or interrupted.

If the consumption input dialog is integrated in the partial upload dialog and if it is not called up when a partial upload is executed or when the entry is interrupted, the system will remember the calculated consumption for all material components of the consumption type "D" and will account for it the next time the consumption input dialog is called up.

Example (assumption: a material component with the input quantity 2 is defined at the operation):

- Call up the partial upload dialog (A_TR): Enter yield 9, scrap 1
- Call up the consumption input dialog (A_VERB)
 - $(9 * 2 + 1 * 2 =) 20$ is proposed as material consumption
 - Cancel the dialog (do not click on OK to confirm)
- Confirm the partial confirmation/upload dialog (A_TR)
 - The yield 9 and the scrap 1 are posted at the operation.
 - The system remembers the calculated consumption of 20
- Call up the partial upload dialog (A_TR) once more: Enter yield of 5
- Call up the consumption input dialog (A_VERB)
 - $(20 + 5 * 2 =) 30$ is now proposed as material consumption
 - Confirm with OK
- Confirm the partial confirmation/upload dialog (A_TR)
 - The yield 5 is posted to the operation
 - The consumption of 30 is posted as a goods movement

If after calling up the consumption input dialog it is canceled and thus closed and the quantity is then modified in the dialog that calls up the function and if then the consumption input dialog is once again called up, then the consumption quantities that were proposed the first time it was called up will still be shown in this dialog. This is why the proposed material consumption will need to be updated by clicking on the "Reset" button as the case may be.

25 Configuration of Discrete Consumption Input

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Summary

Material consumption can be entered at the AIP shop floor terminal while an operation is being interrupted, while an operation is being logged off or when entering a partial confirmation/upload

System configuration

The relevant utilization material must be assigned to the operation in order to discretely record consumption. The meaning of the following fields must be observed in particular (logical consideration). The material components can be assigned in HYDRA manually ([Order management: edit components](#)) or from the upper level system

Material number

The material number uniquely identifies a material. In the component list, a material is only unique in association with a BOM item.

BOM item

The BOM item combined with the material number forms a unique key for a material component at an operation.

Material designation

The material designation is used to describe a material more exactly. It functions as commentary and is displayed in the input dialog.

Material category (type)

The material category controls, among other things, the way a material is technically processed in the HYDRA system. Relating to the material consumption input (not batch-related), the following value is important:

"M" Material component, should be entered in terms of consumption

Material type

The material type is another parameter used for processing control in HYDRA. The material type is used to control whether goods movement created from the material consumption should be uploaded to the ERP system or not. Unless defined otherwise assign the material type SYSTEM here.

To assure that a goods issue is uploaded via the interface, the option "Transfer to interface" must be set at the material type that the utilization material is based on.

When launching HYDRA for the first time or during discrete consumption input, it will need to be coordinated whether or for which material components consumption should be uploaded to the ERP system. If the MPL module is not used, this configuration is to be made during the HYDRA customizing process.

Consumption type

Set the consumption type to "D" (discrete consumption input) for material components that this function is used for to collect material consumption.



It is possible to transfer the consumption type using the relevant component segment of the ERP interface (EIS-ERP). When using the HYINFO interface as a part of the PP-PDC interface, the consumption type is made available by customizing the system accordingly.

The interface necessary to upload material consumption is not a part of this function.

Input quantity

The input quantity is a component quantity needed to manufacture one unit (one piece, for example) of the output material (article/item) being produced. It is used to calculate the theoretical material consumption based on the produced quantity.

Unit

Quantity unit of the material component in which material consumption is recorded.

Dialog configuration

In addition to making the necessary programs available, please also consider the following:

- Activate the dynamic dialog A_VERB, accounting for existing terminal groups in some cases
- Integrate the button used to call up the consumption input dialog in the relevant posting dialogs. The following configurations are possible here, whereas a) can be seen as an alternative to b) and c):
 - a. Integration into the workflow step WF_AA_QUA.
The button is found in the standard dialogs
 - Interrupt OP (A_UN),
 - Log off OP (A_AB), and
 - Partial confirmation/upload (A_TR)
 - in each workflow step in which quantities are entered manually.
 - b. Integration into the workflow dialog WF_AUN_CHK
The button is available in the standard dialog Interrupt OP (A_UN) in the confirmation workflow step.
 - c. Integration into the workflow dialog WF_AAB_CHK
The button is available in the standard dialog Log off OP (A_AB) in the confirmation workflow step.
- Integrating or checking the grid configuration for the dialog A_VERB (ctaipay.ini)

- Optional: Uploading the material consumption as goods movement
 - Determining which material type is transferred to the material component from the ERP system (typically SYSTEM)
 - Setting the flag "Transfer to interface" at the material type. Options:
 - Directly via the master data configuration (only possible, if MPL/ TRT is active)
 - Per SQL to the HYDRA server (hysql -r -):
update hz_typen set we_ext_kz = 'J' where hz_typ = 'SYSTEM';
 - Per SQL via the SQL tester in MOC: update hz_typen set we_ext_kz = 'J' where hz_typ = 'SYSTEM'
 - Per BAPI to the HYDRA server (Observe system!):
hymwb -u9999 -
c"DLG=MATTYP.UPDATE|MATTYP.MATTYP=SYSTEM|MATTYP.WEEXT:GEN
=J|DAT=today|ZEI=now|"
 - Check per SQL: select hz_typ, we_ext_kz from hz_typen
 - Activate the material upload interface (see relevant documentation).

26 AIP - on-screen keyboard decimal symbol configuration

Overview

Menu	All AIP input screens
Transaction code	n/a
Function authorization	n/a

You cannot use the numeric on-screen keyboard (virtual keyboard) to enter decimals, if you configure different decimal symbols in the keyboard settings and the regional settings of the operating system.

Change target cycle for operation 01010100000200010 0010 Turning

Operation 0010 0010 Turning

Order	010101000002	Target qty.	50 PCE
Article	0101H34729-0010	Yield	70 PCE
Handle		Scrap	0 PCE
		Completion	140%

Workplace 01011004 Turning

Current status	PRODUCTION	Part.	2
for	2017-02-22 10:21:43	Current target cycle	5,000 [Sec./cycle]
Duration	9288:23:12	Act. cycle	0,000 [Sec./cycle]

New target cycle [Sec./cycle]

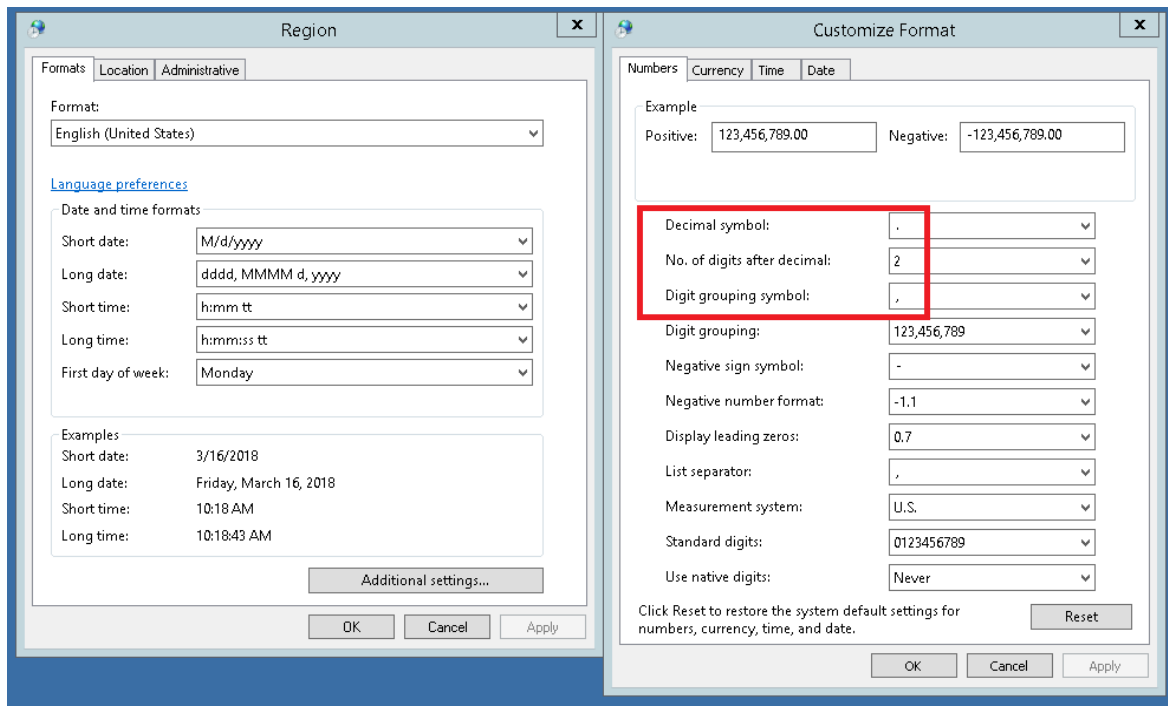
Staff badge number

18-03-16 10:45:09

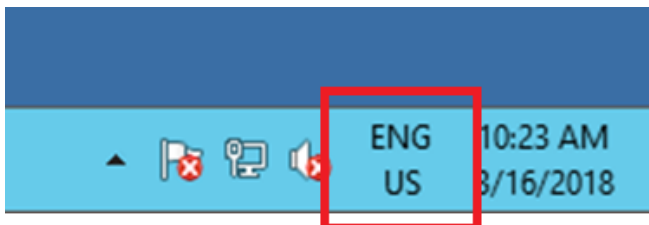
Solution: Keyboard settings and regional settings should match. Example: if you configure a comma "," as the decimal symbol in the regional settings, you also have to select a comma as the decimal separator in the keyboard settings.

Purpose

Every time you start the AIP, the application identifies which decimal symbol is displayed and expected in the input field. The AIP application gets this information from the Windows system setting "region".



Every time you start the on-screen keyboard, the keyboard identifies which decimal symbol is displayed on the numeric keypad. The on-screen keyboard gets this information from the currently selected keyboard settings:



Integration

If you set both configurations correctly (see above), you can use the numeric on-screen keyboard to enter decimals (see the following screenshot):

If the configurations do not match, for example:

- you select the "comma" in the "region" tab of the Windows system settings and
 - you select the "decimal point" in the keyboard settings
- you cannot use the on-screen keyboard to enter the decimal symbol:



AIP

Change target cycle for operation 01010100000200010 0010 Turning

Operation 0010 0010 Turning

Order	010101000002	Target qty.	50 PCE
Article	0101H34729-0010	Yield	70 PCE
Handle		Scrap	0 PCE
		Completion	140%

Workplace 01011004 Turning

Current status	PRODUCTION	Part.	2
for	2017-02-22 10:21:43	Current target cycle	5.000 [Sec./cycle]
Duration	9288:23:12	Act. cycle	0.000 [Sec./cycle]

New target cycle [Sec./cycle]

Staff badge number

mpcv 8.

18-03-16 10:45:09